

As Per NEP 2020

University of Mumbai



Syllabus for Major Vertical – 1, 4 & 5

Name of the Programme – B.E. (<u>Automation and Robotics</u>)		
Faculty of <u>Engineering</u>		
Board of Studies in Mechanical Engineering		
U.G. Third Year Programme	Exit Degree	B.Sc. Tech Engineering- <u>Automation and Robotics</u>
Semester	V & VI	
From the Academic Year	2026-27	

University of Mumbai



(As per NEP 2020)

Sr. No.	Heading	Particulars
1	Title of program O: _____	B.E. (<u>Automation and Robotics</u>)
2	Exit Degree	B.Sc. Tech Engineering- <u>Automation and Robotics</u>
3	Scheme of Examination R: _____	NEP 40% Internal 60% External, Semester End Examination Individual Passing in Internal and External Examination
4	Standards of Passing R: _____	40%
5	Credit Structure R: _____	Attached herewith
6	Semesters	Sem. V & VI
7	Program Academic Level	5.5
8	Pattern	Semester
9	Status	New
10	To be implemented from Academic Year	2026-27

Sd/-
Dr. S M Khot
BoS-Chairman-Mechanical Engineering
Faculty of Technology

Sd/-
Dr. Deven Shah
Associate Dean
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Prof. Shivram S. Garje
Dean
Faculty of Science & Technology

Preamble

To meet the challenge of ensuring excellence and NEP 2020 policy in engineering education, the issue of quality needs to be addressed, debated, and taken forward systematically. Accreditation is the principal means of quality assurance in higher education. The major emphasis of the accreditation process is to measure the outcomes of the program that is being accredited. In line with this Faculty of Science and Technology (in particular Engineering) of the University of Mumbai has taken the lead in incorporating the philosophy of NEP 2020 education in the process of curriculum development.

The **third-year** engineering course consists of a core and elective training program to impart scientific and logical thinking training to learners in general, with a choice of course selection from the program core course, program elective course, multidisciplinary minor, IKS, PCE and vocational skill-enhanced course. Simultaneously, the objectives of NEP 2020 demand nurturing the core program, elective program, communication skills and skills required for the **Automation and Robotics** Branch of engineering in the learner. Keeping this in view, a pool of courses is offered in Core Courses covering fundamentals required to understand core and modern engineering practices and emerging trends. Elective courses are offered to the learners to understand the concepts of emerging technology. This will help with the holistic development of learners through interdisciplinary skills development and mobility among other departments as real-time solutions. Considering the shift in pedagogy and the convenience of a stress-free learning process, a choice-based subject pool is offered in the coursework under the heads of **Automation and Robotics** in Engineering for open electives and multidisciplinary minor courses in SEM V and VI. Essentially, to give a glimpse of trends in the industry under vocational and enhanced skill practices, the pool is offered to nurture and develop creative skills in contemporary industrial practices. Criteria met in the structure is the opportunity for learners to choose the course of their interest in all disciplines.

Program Core Course and Program Elective Courses Cover **Automation and Robotics** core and elective courses. Also, MDM is where a pool of subjects is given for selection. In addition to this learner will have IKS and PCE courses. Considering the present scenario, diverse choices need to be made available to fulfil the expectation of a learner to aspire for a career in the field of current trends of Technology and interdisciplinary research. The IKS course will be program-specific, which can be achieved in undergraduate training by giving an objective viewpoint to the learning technology, from ascent time and how it was learned and the process and transitioning a learner from a rote learner to a creative professional. For this purpose, PCE is introduced in the First Semester to orient a communication skills to become skilled professionals for placement and higher education entrance exams. Considering the NEP-2020 structure of award of Certificate after U.G. First Year, U. G. Diploma after the Second Year and B.Sc. Tech Engineering after the Third Year at multiple exit-point pools of Vocational skills, is arranged to give exposure to the current Industry practices.

The faculty resolved that course objectives and course outcomes are to be clearly defined for every core and elective course so that all faculty members in affiliated higher education institutes understand the depth and approach of the course to be taught to the learners, which will enhance the learners' learning process. NEP 2020 grading system enables a much-needed shift in focus from teacher-centric to continuous-based learner-centric education since the workload estimated is based on the investment of time in learning and not in teaching. It also focuses on the holistic development of the learners through mobility and continuous evaluation, which will enhance the quality of education. Credit assignment for courses is based on a 15-week teaching-learning process for NEP 2020; however, the content of courses is to be taught in 12-13 weeks, and the remaining 2-3 weeks are to be utilized for revision, self-learning, tutorial, guest lectures, coverage of content beyond the syllabus, etc.

There was a concern that in the present system, the **third-year** syllabus must not be heavily loaded for the learner and it is of utmost importance that the learner entering into the third year of an engineering course should feel at ease by lowering the burden of syllabus and credits. This is necessary for a learner to get accustomed to the new environment of a college and to create a bond between the teacher and the learner. The present curriculum will be implemented for the **Third Year** of Engineering from the academic year **2026-27**. Subsequently, this system will be carried forward for Final Year Engineering in the academic years 2027-28, respectively.

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B.Sc. Tech Engineering- Automation and Robotics.

Credit Structure (Sem. V & VI)

	R: _____ C									
Level	Semester	Major		Minor	OE	VSC, SEC (VSEC)	AEC, VEC, IKS	OJT, FP, CEP, CC,RP	Cum. Cr. / Sem.	Degree / Cum. Cr.
		Mandatory	Electives							
5.5	V	PCC501:3 PCC502:3 PCC503:3 PCL501: 1 PCL502:1 PCL503:1	PEC501:3 PEL501:1	MDC:3 MDL:1	OE:2	--	IKS: 2	--	24	B.Sc. Tech Enginee ring 48
	R: _____ D									
	VI	PCC601:3 PCC602:3 PCL601:1 PCL602:1	PEC601:3 PEC602:3 PEL601:1 PEL602:1	MDC: 3 MDL:1	--	VSEC:2	AEC: 2	--	24	
	Cum Cr.	20	12	8	2	2	2+2	--	48	
Exit option: Award of BSc. Tech Engineering with 138 credits and additional 4 credits core one theory subject with 3 credits and one lab with 1 credit from one final year from where they want to take Exit degree. Along with theory and practical course student must compulsory do internship for one month or 160 hours which internship is equal to 4 credits.										

[Abbreviation - OE – Open Electives, VSC – Vocation Skill Course, SEC – Skill Enhancement Course, (VSEC), AEC – Ability Enhancement Course, VEC – Value Education Course, IKS – Indian Knowledge System, OJT – on Job Training, FP – Field Project, CEP – Continuing Education Program, CC – Co-Curricular, RP – Research Project]

T.E. Automation And Robotics Scheme

SEMESTER V

Course Code	Course Description	Teaching Scheme (Contact Hours)			Credit Assigned			
		Theory	Practical	Tutorial	Theory	Tutorial	Practical	Total Credits
2035111	Power Electronics and Drives	3	--	--	3	--	--	3
2035112	Robot Kinematics and Dynamics	3	--	--	3	--	--	3
2035113	Digital Signal Processing	3	--	--	3	--	--	3
2035114X	PEC-1	3	--	--	3	--	--	3
MDC501	Multidisciplinary minor	3	--	--	3	--	--	3
2035115	Power Electronics and Drives Lab	--	2	--	--	--	1	1
2035116	Robot Kinematics and Dynamics Lab	--	2	--	--	--	1	1
2035117	Digital Signal Processing Lab	--	2	--	--	--	1	1
2035118X	PEL-1 Lab	--	2	--	--	--	1	1
MDL501	Multidisciplinary minor Lab	--	2	--	--	--	1	1
OECL501	To be taken from the bucket provided by the University from other Faculty	2	--	---	2	--	--	2
2035511	Indian Knowledge System-II	--	2*+2	--	--	--	2	2
Total		17	14	00	17	00	07	24

*Two hours of practical class to be conducted for full class as demo/discussion.

Theory / Tutorial 1 credit for 1 hour and Practical 1 credit for 2 hours

Institute shall offer a course for Open Elective from Science/Commerce/Management stream bucket provided by the University of Mumbai.

@ Institute shall offer a course for MDM from other Engineering Boards Group.

Course Code	Program Elective Course (PEC-1)
20351141	Data Structures and Algorithms
20351142	Optimization Techniques
20351143	Advanced Sensors & Actuators
Lab Code	Program Elective Course PEL-1 (Lab)
20351181	Data Structures and Algorithms Lab
20351182	Optimization Techniques Lab
20351183	Advanced Sensors & Actuators Lab

Course Code	Course Description	Examination scheme							
		Internal Assessment Test (IAT)			End Sem. Exam Marks	End Sem. Exam Duration (Hrs)	Term Work (Tw)	Oral & Pract.	Total
		IAT-I	IAT-II	Total (IAT-I) + (IAT-II)					
2035111	Power Electronics and Drives	20	20	40	60	2	--	--	100
2035112	Robot Kinematics and Dynamics	20	20	40	60	2	--	--	100
2035113	Digital Signal Processing	20	20	40	60	2	--	--	100
2035114X	PEC-1	20	20	40	60	2	--	--	100
MDC501	Multidisciplinary minor	20	20	40	60	2	--	--	100
2035115	Power Electronics and Drives Lab	--	--	--	--	--	25	25	50
2035116	Robot Kinematics and Dynamics Lab	--	--	--	--	--	25	--	25
2035117	Digital Signal Processing Lab	--	--	--	--	--	25	25	50
2035118X	PEL-1 Lab	--	--	--	--	--	25	25	50
MDL501	Multi-disciplinary minor Lab	--	--	--	--	--	25	25	50
OECL501	To be taken from the bucket provided by the University from other Faculty	--	--	--	--	--	50	--	50
2035511	Indian Knowledge System-II	--	--	--	--	--	50	--	50
Total		100	100	200	300	10	225	100	825

SEMESTER VI

Course Code	Course Description	Teaching Scheme (Contact Hours)			Credit Assigned			
		Theory	Practical	Tutorial	Theory	Tutorial	Practical	Total Credits
2036111	Robotic Control System	3	--	--	3	--	--	3
2036112	Process Instrumentation and Control	3	--	--	3	--	--	3
2036113X	PEC-2	3	--	--	3	--	--	3
2036114X	PEC-3	3	--	--	3	--	--	3
MDC601	Multi-disciplinary minor	3	--	--	3	--	--	3
2036115	Robotic Control System Lab	--	2	--	--	--	1	1
2036116	Process Instrumentation and Control Lab	--	2	--	--	--	1	1
2036117X	PEL-2 Lab	--	2	--	--	--	1	1
2036118X	PEL-3 Lab	--	2	--	--	--	1	1
2036411	3D Printing Lab (Mini Project)	--	2*+2	--	--	--	2	2
2036511	Professional Skill	--	2*+2	--	--	--	2	2
MDL601	Multi-disciplinary minor Lab	--	2	--	--	--	1	1
Total		15	18	00	15	00	09	24

*Two hours of practical class to be conducted for full class as demo/discussion.

Theory / Tutorial 1 credit for 1 hour and Practical 1 credit for 2 hours

Institute shall offer a course for Open Elective from Science/Commerce/Management stream bucket provided by the University of Mumbai.

@ Institute shall offer a course for MDM from other Engineering Boards Group.

Course Code	Program Elective Course (PEC-2)
20361131	Database Management System (DBMS)
20361132	Industrial Robotics
20361133	Internet of Things (IOT)
Course Code	Program Elective Course (PEC-3)
20361141	Biomedical Instrumentation
20361142	Micro Electro Mechanical System
20361141	Hydraulics and Pneumatics
Lab Code	Program Elective Course PEL-2 (Lab)
20361171	Database Management System (DBMS)
20361172	Industrial Robotics
20361173	Internet of Things (IOT)
Lab Code	Program Elective Course PEL-3 (Lab)
20361181	Biomedical Instrumentation
20361182	Micro Electro Mechanical System
20361183	Hydraulics and Pneumatics

Course Code	Course Description	Examination scheme							
		Internal Assessment Test (IAT)			End Sem. Exam Marks	End Sem. Exam Duration (Hrs)	Term Work (Tw)	Oral & Pract.	Total
		IAT-I	IAT-II	Total (IAT-I) + IAT-II)					
2036111	Robotic Control System	20	20	40	60	2	--	--	100
2036112	Process Instrumentation and Control	20	20	40	60	2	--	--	100
2036113X	PEC-2	20	20	40	60	2	--	--	100
2036114X	PEC-3	20	20	40	60	2	--	--	100
MDC601	Multi-disciplinary minor	20	20	40	60	2	--	--	100
2036115	Robotic Control System Lab	--	--	--	--	--	25	25	50
2036116	Process Instrumentation and Control Lab	--	--	--	--	--	25	25	50
2036117X	PEL-2 Lab	--	--	--	--	--	25	25	50
2036118X	PEL-3 Lab	--	--	--	--	--	25	25	50
2036411	3D Printing Lab (Mini Project)	--	--	--	--	--	50	25	75
2036511	Professional Skill	--	--	--	--	--	25	-	25
MDL601	Multi-disciplinary minor Lab	--	--	--	--	--	25	25	50
Total		100	100	200	300	10	200	150	850

Semester - V

Vertical – 1

Major

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2035111	Power Electronics and Drives	3	-	-	3	-	-	3

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
2035111	Power Electronics and Drives	20	20	40	60	2	--	1	100

Rationale:

Power Electronics and Drives equips students with knowledge of efficient power conversion and motor control. It enables understanding of converters and electric drives used in industries, automation, and electric vehicles, helping design energy-efficient, reliable systems essential for modern technological and sustainable engineering applications.

Course Objectives:

1. To study the basics of semiconductor devices with the help of operation, characteristics, and the methods used to choose appropriate power devices.
2. To Acquire and compare the converter circuits such as rectifiers, choppers, and inverters.
3. To study power electronics based rectifiers, inverters and choppers.
4. To understand principles of electric drives and speed control methods.
5. To apply power electronics concepts in automation and robotics systems.
6. To develop analytical skills for industrial drive applications.

Course Outcomes:

1. Apply knowledge of basic operations to analyze and compare the characteristics of various semiconductor devices.
2. Design of triggering, commutation and protection circuits for SCRs.
3. Analyze controlled rectifiers circuits with various loads.
4. Analyze different types of operation of choppers and inverters.
5. Analyze fundamentals of electric drives and apply speed control methods for DC drives.
6. Apply various auxiliary circuits and explain the working and applications of electrical drives.

DETAILED SYLLABUS:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Basic knowledge of electrical circuits, semiconductor devices (diodes, transistors), and AC/DC fundamentals is required.	0	--
I	Thyristors and Power semiconductor devices	1.1 Power Diode, construction and operation of SCR, Static V-I characteristics, gate control, Two transistor Analogy, 1.2 Basic principle of structure, operation, ratings, V-I characteristics and applications of: TRIAC, DIAC, power MOSFET, IGBT, Comparison of Power BJT, SCR, PMOSFET & IGBT,	6	CO 1
II	Triggering, commutation and Protection Circuits	2.1 turn - ON methods of SCR, firing circuits (using R, RC, UJT, Ramp and pedestal, inverse cosine), 2.2 Define Commutation methods of SCR, 2.3 Protection of SCR	6	CO 2
III	Controlled Rectifiers	3.1 Single phase half-wave controlled rectifiers with R, RL load, effect of free-wheeling diode and applications 3.2 Single phase full-wave controlled rectifiers (center-tapped, bridge configurations) with R, RL load, effect of freewheeling diode and effect of source inductance. 3.3 Three phase semi converter and full converter with R load, Calculation of performance parameters, input performance parameters (input power factor, input displacement factor (DF), input current distortion factors (CDF), input current harmonic factor (HF/THD), Crest Factor (CF)), output performance parameters.	7	CO 3
IV	Inverter:	4.1 Introduction, principle of operation, performance parameters of Single phase half/full bridge Inverters with R and R-L load, 4.2 Introduction, principle of operation, performance parameters of Three phase VSI (120° and 180° conduction mode), 4.3 Voltage control of single phase inverters- PWM techniques, Basics of Space vector modulation, Single phase current source inverters (CSI), Cyclo-converter, comparison of VSI, PWM and CSI.	6	CO 4

V	DC-DC converters	5.1 Basic principle of step up and step down DC-DC converters, duty cycle control, PWM, applications. DC-DC switching mode regulators: Buck, Boost, Buck-Boost, chuk Regulators (CCM mode only) 5.2 Voltage commutated, current commutated and load commutated DC-DC converters 5.3 Applications in SMPS, Battery charging systems	7	CO 5
VI	Drives Circuits and Power Electronics Applications	6.1 Introductions to drives, DC drives and AC drives 6.2 Load and Speed- torque characteristics, Induction motor drives (V/f control), 6.3 Robotic applications: Robotic Arm Drives and Control, Drive system for articulated arms: stepper, servo, BLDC motors. Motion Control in Automated System, Integration of electrical drives PLCs for automation system, Case studies of Applications of Electrical Drives in Robotics and Automation.	7	CO 6

Text Books:

1. M. H. Rashid, Power Electronics: Circuits, Devices, and Applications, Pearson Education, 2009.
2. N. Mohan and T. M. Undeland, Power Electronics: Converters, Applications and Design, John Wiley & Sons, 2007.
3. R.W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer Science & Business Media, 2007.
4. P.C Sen., Modern Power Electronics, Wheeler publishing Company, 1st Edition, 2005
5. Alok Jain, Power Electronics: Devices, Circuits and Matlab Simulations, Penram Int. 2010
6. G. K. Dubey, "Fundamentals of Electric Drives" 2 nd Edition, Narosa Publications, 2002.
7. Vedam Subrahmanyam, "Electric Drives: Concepts & Applications", 2nd Edition, Tata McGraw-Hill Education, 2010.

References:

1. C.W. Landers, Power Electronics, McGraw Hill, 1993
2. Ashfaq Ahmed, Power Electronics for Technology, Pearson, 1998
3. Joseph Vithayathil, Power Electronics, Tata McGraw hill, 1995.
4. P. Friedrichs, T. Kimoto, L. Ley and G. Pensl, Silicon Carbide, Volume 2: Power Devices and Sensors, Wiley Publications, 2011.
5. Dokić, Branko L. and Blanuša, Branko, Power Electronics Converters and Regulators Springer International Publishing, 2015
6. L. Umanand, Power Electronics: Essentials and Applications, Wiley India, 2009.
7. B. Jayant Baliga, Silicon Carbide Power Devices, World Scientific, 2005

Online References:

Sr. No.	Website Name
1.	Advance Power Electronics And Control – Prof. Avik Bhattacharya (IIT Roorkee) https://nptel.ac.in/courses/108/107/108107128/
2.	Power Electronics By Prof. G. Bhuvaneshwari (IIT Delhi) https://swayam.gov.in/nd1_noc20_ee97/preview

Assessment:**Internal Assessment (IA) for 40 marks:**

- IA will consist of Two Compulsory Internal Assessment Tests of 20 marks each. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 60% to 50% of syllabus content must be covered in Second IA Test

End Semester Examination for 60 Marks

Question paper format

- Question Paper will comprise of a total of six questions each carrying 15 marks Q.1 will be compulsory and should cover maximum contents of the syllabus
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of Four questions needs to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2035112	Robot Kinematics and Dynamics	3	-	-	3	-	-	3

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
2035112	Robot Kinematics and Dynamics	20	20	40	60	2	--	--	100

Rational:

Kinematics deals with the geometry of motion to determine a robot's position, orientation, and movement without considering forces, which is essential for tasks like path planning and motion Simulation. Dynamics builds on this by adding the forces and torques involved, allowing engineers to Understand the relationship between motor effort and the resulting motion, enabling accurate control And the ability to calculate necessary joint torques for specific tasks, such as those in industrial or medical Applications

Course Objectives:

1. To familiarize the types of mechanisms for various robotic applications.
2. To analyze the basic Geometrical and Algebraic approach to solve forward kinematics of serial manipulator
3. To apply knowledge about advanced forward kinematics of serial manipulators.
4. To evaluate the inverse kinematics of various serial manipulators.
5. To familiarize the design of robot manipulators based on dynamic analysis
6. To analyze balancing and vibrations

Course Outcomes:

1. Apply mechanisms in robotic applications.
2. Analyze forward kinematic models for planar and spatial robot manipulators.
3. Apply forward kinematic models to multi-DOF robot manipulators.
4. Apply inverse kinematic models to multi-DOF robot manipulators.
5. Apply design procedures for mechanical grippers based on their types and mechanisms.
6. Apply and analyze dynamic characteristics of robot manipulators.

DETAILED SYLLABUS:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Robot introduction, robot terminology, degree of freedom, a strong foundation in undergraduate mathematics , including calculus, linear algebra, and differential equations, along with engineering mechanics .	0	-
I	Robot Mechanisms	Kinematic Link, Types of links, Kinematics pair, Types of constrained motion, Classification of Kinematics pairs, Kinematics chain, Degrees of freedom of mechanisms, Inversion of mechanism, Analysis of mechanisms such as Gear trains, cams and followers, belt drives, four bar mechanism, slider crank mechanism etc.	06	CO1
II	Forward Kinematics - Geometrical And Algebraic Approach	Need for forward and Inverse Kinematics Equation – Parameters in Design and Control – Methods of forward and inverse kinematics- Geometrical and Algebraic Approach in Forward Kinematics Solution, 1 DOF - 2 DOF Planar Robot (2P and 2R); 3DOF 2RP Spatial Robot.	06	CO2
III	Forward Kinematic Modeling – Denavit-Hartenberg (Dh) Approach	Unit Circle Trigonometry - Translation Matrix - Rotation matrix, Euler Angles - Quaternion Fundamental - Dot and Cross Products - Frames and Joint Coordinates - Homogeneous Transformation - D-H and Modified D- H Convention and Procedures – Forward kinematics Solution using D-H Convention: 3 DOF wrist, RR Planar, 3 DOF RRP, Cartesian, Cylindrical, Spherical, SCARA and Articulated 3 DOF robots - 3 DOF robot with wrist.	06	CO3
IV	Inverse Kinematics Modeling	Introduction to inverse kinematics -Issues in inverse kinematics - Inverse kinematics of 2 DOF Planar robot - 2 and 3 DOF planar and Spatial robot - Tool configuration - Inverse kinematics of 3 axis robot and 6 axis Robot - Inverse kinematics Computation- Closed loop solution	07	CO4
V	Robot End Effectors	Types of end effectors, mechanical grippers, vacuum grippers, magnetic grippers, adhesive grippers, tools. Force analysis, the robot end effectors interface, considerations in gripper selection and design. Robot Arm Dynamics Robot dynamics – Rigid body dynamics, Newton-	07	CO5

		Euler formation, Lagrange-Euler, formation, generalized D'Alembert equations of motion.		
VI	Balancing And Vibrations	Static and Dynamic balancing, balancing of revolving and reciprocating masses, Balancing machines, free vibrations, Equations of motion, natural Frequency, Damped Vibration, bending critical speed of the simple shaft.	07	CO6

TEXT BOOKS:

1. Mikell P. Groover, "Industrial Robotics", McGraw Hill, 2nd edition, 2012.
2. John J. Craig, "Introduction to Robotics", 3rd Edition, Addison Wesley, ISE 2008.
3. Lynch, Kevin M., and Frank C. Park. Modern Robotics: Mechanics, Planning, and Control 1st ed. Cambridge University Press, 2017.

REFERENCES:

1. S K Saha, Introduction to Robotics, Tata McGraw-Hill, Second Edition, 2017.
2. Mikell P. Groover, "Industrial Robotics", McGraw Hill, 2nd edition, 2017.
3. Arthor Critchlow, "Introduction to Robotics", 1st edition, Macmillan, 2009.
4. Mohsen Shahinpoor, "A Robot Engineering Text Book", 1st edition, Harper and Row, 2004.
5. Deb S.R., "Robotics Technology and Flexible Automation", 2nd edition, Tata McGraw - Hill Publis
6. J. Srinivas, R. V. Dukkipati, K., "Robotics: Control and Programming", Narosa Publishing House, 2009.

Online References:

Sr. No.	NPTEL Link
1.	https://nptel.ac.in/courses/112105249
2.	https://nptel.ac.in/courses/11210523
3.	https://nptel.ac.in/courses/112108298

Assessment:

Internal Assessment (IA) for 40 marks:

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- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of Four questions needs to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2035113	Digital Signal Processing	3	-	-	3	-	-	3

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs.)			
		Test 1	Test 2	Total					
2035113	Digital Signal Processing	20	20	40	60	2	--	--	100

Rationale:

Digital Signal Processing provides the foundation for analyzing, modifying, and interpreting real-world signals such as audio, images, and sensor data. It enables efficient, accurate, and flexible processing using digital systems, making it essential in modern technologies like communication, biomedical devices, and multimedia. The subject equips students with skills to design and implement algorithms crucial for advanced engineering applications.

Course Objectives:

1. To make the students familiar with the fundamentals of digital signal processing.
2. To make the students understand the need of frequency domain analysis of Digital Signal.
3. To make the students understand the transforms used in Digital Signal Processing.
4. To make the students familiarize with the design techniques and performance analysis of digital filters
5. To introduce digital signal processors and its architecture.
6. To make the students familiar use of DSP in different applications of automation and robotics

Course Outcomes:

1. Apply the concept of Discrete Time Signal and Discrete Time Systems.
2. Classify and analyze discrete time signals and systems.
3. Implement Digital Signal Transform techniques DTFT, DFT and FFT.
4. Design IIR digital filters to meet arbitrary specifications and Develop algorithms for implementation.
5. Design FIR digital filters to meet arbitrary specifications and Develop algorithms for implementation.
6. Apply signal processing techniques and digital signal processors to analyze various applications.

DETAILED SYLLABUS:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Understanding of: Continuous-time and discrete-time signals, Linear time-invariant (LTI) systems, Convolution, Fourier series and Fourier transform, Laplace and Z-transforms, Basic differentiation & integration, Solutions of differential equations, Complex numbers and Euler's formula etc.	0	-
I	Discrete-Time Signal and Discrete-Time Systems	Introduction to Digital Signal Processing, Sampling and Reconstruction, Standard DT Signals, Concept of Digital Frequency, Representation of DT signal using Standard DT Signals, Signal Manipulations-shifting, reversal, scaling, addition, multiplication. Classification of Discrete-Time Signals, Classification of Discrete- Systems, LTI system, Impulse Response. Linear Convolution, Circular Convolution- Emphasis on graphical method, linear convolution using Circular Convolution. Software simulation - Impulse Response, Step Response, convolution, Correlation.	08	CO1
II	Frequency Domain Analysis using DTFT and Z Transform	Introduction to DTFT. Properties of DTFT. Z transform - definition, properties of unilateral and bilateral Z Transform, Z transform of standard signals, ROC, poles and zeros of transfer function, Inverse Z transform. Analysis and characterization of LTI system using Z transform, impulse and step response, causality, stability, stability of causal system.	07	CO2
III	Discrete Fourier Transform and Fast Fourier Transform	DFT, Relation between DFT and DTFT, IDFT. Properties of DFT, circular convolution of sequences using DFT. Fast Fourier transforms (FFT), Radix-2 decimation in time and decimation in frequency FFT algorithms, inverse FFT.	06	CO3
IV	IIR Digital Filters	Comparison of IIR and FIR filters, Types of IIR Filters, Analog filter approximations: Butterworth, Chebyshev I and II. Mapping of S-plane to Z-plane, impulse invariance method, bilinear transformation method, Design of IIR digital filters from analog filters with examples, Software simulation – Design of IIR Filters. Analog and digital frequency transformations.	07	CO4

V	FIR Digital Filters	Characteristics of FIR digital filters, Minimum Phase, Maximum Phase, Mixed Phase and Linear Phase Filters Frequency response, location of the zero of linear phase FIR filters. Design of FIR filters using window techniques - Rectangular, Hamming, Hanning, Blackman, Bartlett, Software simulation – Design of FIR Filters.	07	CO5
VI	DSP Processors and Applications	General purpose digital signal processors, DSP processor architecture, Selecting digital signal processors, Special purpose DSP hardware. Applications of DSP: Radar Signal Processing and Speech Processing.	04	CO6

Text Books:

1. Tarun Kumar Rawat, “Digital Signal Processing”, Oxford University Press, 2015
2. S Salivahanan, A Vallavaraj, C Gnanapriya. “Digital Signal Processing” – TMH, 2007
3. Emmanuel C. Ifeakor, Barrie W. Jervis, “Digital Signal Processing”, A Practical Approach by, Pearson Education – Second edition

References:

1. Proakis J., Manolakis D., Digital Signal Processing, 4th Edition, Pearson Education
2. Sanjit K. Mitra, “Digital Signal Processing – A Computer Based Approach”, edition 4e McGraw Hill Education (India) Private Limited.
3. Analog And Digital Communication, Special Indian Edition, Schaum S Outline Series

Online References:

Sr. No.	Website Name
1.	https://nptel.ac.in/courses/108104100
2.	https://ocw.mit.edu/courses/res-6-007-signals-and-systems-spring-2011/pages/lecture-no
3.	https://www.iitg.ac.in/cseweb/vlab/Signal-SystemLab/signalsystem/Signals%20and%20their%20properties(simulator).html

Assessment:

Internal Assessment (IA) for 40 marks:

- IA will consist of Two Compulsory Internal Assessment Tests of 20 marks each. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 60% to 50% of syllabus content must be covered in Second IA Test

End Semester Examination for 60 Marks

Question paper format

- Question Paper will comprise of a total of six questions each carrying 15 marks Q.1 will be compulsory and should cover maximum contents of the syllabus
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of Four questions needs to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20351141	Data Structures and Algorithms	3	-	-	3	-	-	3

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
20351141	Data Structures and Algorithms	20	20	40	60	2	--	--	100

Rationale:

An algorithm is a step-by-step set of instructions for solving a problem. Algorithms are fundamental to computer science, and knowing how to effectively store and manage data is crucial for designing efficient algorithms. Data structures provide a way of organizing data so that it can be processed, stored, and retrieved quickly and effectively. This course will cover a variety of data structures and their applications in algorithms.

Course Objectives:

1. To learn mathematical background for Analysis of Algorithms
2. To learn and use various advanced data structures
3. To learn and use greedy approach to solve problems
4. To learn and use dynamic programming algorithms
5. To learn and use backtracking and branch and bound approach.
6. To learn and use pattern matching algorithms.

Course Outcomes:

1. Apply techniques to analyze the time and space complexity of algorithms.
2. Apply the Divide and Conquer strategy and analyze its computational complexity.
3. Apply the Greedy algorithm approach and analyze its efficiency.
4. Analyze solutions using Dynamic Programming and examine its complexity.
5. Apply and analyze Backtracking and Branch and Bound techniques.
6. Apply String matching algorithms for pattern searching.

DETAILED SYLLABUS:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Overview of Data Structures: Revision of basic data structures (arrays)	00	
I	Introduction to Data Structures and Algorithms	Fundamentals of the analysis of algorithms: Time and Space complexity, Asymptotic notation, , Big- Oh, Omega Theta notation Growth of function Mathematical background for algorithm analysis. Recurrence Relations: Analysis of selection sort, insertion sort.	05	CO1
II	Divide and Conquer Approach	General method, Merge sort, Quick sort, Analysis of Binary search. Self-learning Topics: Finding minimum and maximum algorithms and their Analysis, Strassen's Algorithm, real life applications of all algorithms related to Automation and Robotics	05	CO2
III	Greedy Method Approach	General Method, Single source shortest path: Dijkstra Algorithm Fractional Knapsack problem, Minimum cost spanning trees: Kruskal and Prim's algorithms Self-learning Topics: Job sequencing with deadlines, real life applications of all algorithms related to Automation and Robotics	07	CO3
IV	Dynamic Programming Approach	General Method, Multistage graphs, All pair shortest path: Floyd Warshall Algorithm, 0/1 knapsack Problem, Travelling Salesperson problem, Longest Common Subsequence. Self-learning Topics: Bellman Ford Algorithm, real life applications of all algorithms related to Automation and Robotics	08	CO4
V	Backtracking and Branch and bound	General Method, Backtracking: N-queen problem, Sum of subsets, Graph Coloring. Branch and Bound: Travelling Salesperson Problem, 15 Puzzle problem Self-learning Topics: Real life applications of all algorithms related to Automation and Robotics	07	CO5
VI	String Matching Algorithms	The Naïve string-matching algorithm, The Rabin Karp algorithm, The Knuth-Morris-Pratt algorithm Self-learning Topics: Real life applications of all algorithms related to Automation and Robotics	07	CO6

Text Books:

1. T. H. Cormen, C.E. Leiserson, R. L. Rivest, and C. Stein, "Introduction to algorithms", 2nd Edition, PHI Publication 2005
2. Ellis Horowitz, Sartaj Sahni, S. Rajsekaran. "Fundamentals of computer algorithms" University Press.
3. Jon Kleinberg and Eva Tardos, "Algorithm Design", Pearson Education.
4. Mark Allen Weiss, "Data Structures and Algorithm Analysis in C++", Pearson Education.

References:

1. Sanjoy Dasgupta, Christos Papadimitriou, Umesh Vazirani, "Algorithms", Tata McGraw- Hill Edition.
2. S. K. Basu, "Design Methods and Analysis of Algorithm", PHI
3. Anany Levitin, "Introduction to the Design and Analysis of Algorithms", Pearson Education.
4. Steven S. Skiena, "The Algorithm Design Manual", Springer

Online References:

Sr. No.	Website Name
1.	https://nptel.ac.in/courses/106/106/106106131/
2.	https://swayam.gov.in/nd1_noc19_cs47/preview
3.	https://www.coursera.org/specializations/algorithms

Assessment:**Internal Assessment (IA) for 40 marks:**

- IA will consist of Two Compulsory Internal Assessment Tests of 20 marks each. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 60% to 50% of syllabus content must be covered in Second IA Test

End Semester Examination for 60 Marks**Question paper format**

- Question Paper will comprise of a total of six questions each carrying 15 marks Q.1 will be compulsory and should cover maximum contents of the syllabus
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of Four questions needs to be answered

Course Code	Course Name	Theory					Term work	Pract. / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
20351142	Optimization Techniques	20	20	40	60	2	--	--	100

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20351142	Optimization Techniques	3	-	-	3	-	-	3

Rationale:

Optimization Techniques is essential for Automation and Robotics engineering as it enables students to develop mathematical and analytical skills to find the best possible solutions for complex engineering problems. The course covers linear, nonlinear, and integer programming, along with modern optimization and decision-making methods, helping students improve system performance, efficiency, and reliability. It also introduces simulation and design of experiments, which are important for real-world applications in automation, manufacturing, and robotic systems.

Course Objectives:

1. To understand and formulate optimization problems in engineering applications.
2. To apply linear and integer programming techniques for decision-making problems.
3. To analyze and solve nonlinear optimization problems using classical methods.
4. To develop skills in multi-objective and multi-criteria decision-making.
5. To apply simulation techniques for real-world systems.
6. To use robust design and DOE techniques for process optimization.

Course Outcomes:

1. Apply basic concepts and classification of optimization problems.
2. Formulate and solve linear programming problems using various methods.
3. Solve integer and nonlinear optimization problems.
4. Apply multi-objective optimization techniques.
5. Apply multi-criteria decision-making methods to analyze engineering problems.
6. Design experiments and apply robust design techniques.

DETAILED SYLLABUS:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Prerequisites: Mathematics: calculus, algebra, and linear algebra, LPP, Linear algebra, Basic Statistics: Probability Theory Statistical Inference ,Data Interpretation, Mathematical Programming Fundamentals: Basic Optimization ,Basic Programming Skills: Python, MATLAB, Decision-Making and Problem Solving.	00	-
I	Basic Concepts:	Statement of the Optimization Problem, Basic Definitions, Optimality Criteria for Unconstrained Optimization, Optimality Criteria for Constrained Optimization, Engineering Application of Optimization, Classification of Optimization Problems. Classical Optimization Techniques: Single variable optimizations	08	CO1
II	Linear Programming Problem:	Formulation, Simplex method, Big M Method, Two Phase, Primal to Dual, Dual Simplex method, Sensitivity Analysis and applications of LP Transportation and Assignment Models.	07	CO2
III	Integer Programming Model:	Gomory's cutting plane method, Branch & Bound Technique. Non L.P. Model: Lagrangian method & Kuhn tucker Method, Newton's method. Discrete Event Simulation: Generation of Random Variable, Simulation Processes, Monte-Carlo Technique.	07	CO3
IV	Multi Objective Decision making (MODM) Methods:	Introduction to Multi objective optimization, Traditional Techniques such as, quadratic programming, geometric programming, Numerical on goal programming and dynamic programming. Introduction to Non-traditional optimization Techniques such as Genetic Algorithm, particle swarm, genetic algorithms, simulated annealing and Techniques based on Neural network & Fuzziness (Only concepts)	07	CO4
V	Multi Criterion Decision-making (MCDM) Methods:	Introduction to multi criterion optimization, Simple Additive Weighting (SAW) Method Weighted Product Method (WPM), Analytic Network Process (ANP), Analytic Hierarchy Process (AHP) Method, TOPSIS Method PROMETHEE	08	CO5

VI	Robust Design Methods: DOE and Taguchi techniques Full Factorial Design:	Introduction to Design of Experiments (DOE), Taguchi Techniques for Robust Design, Full Factorial Design, ANOVA (Analysis of Variance), Fractional Factorial Design (2^{n-1} , 2^{n-2}), Use of Software: MATLAB / Minitab / Design Expert	03	CO6
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Text Book:

1. S. S. Rao, "Engineering Optimization - Theory and Practice", John Wiley and Sons Inc.
2. Ranjan Ganguli, "Engineering Optimization - A Modern Approach" Universities Press
3. Pablo Pedregal, "Introduction to Optimization", Springer
4. L. C. Jhamb, "Quantitative Techniques Vol. 1 and 2", Everest Pub. House

Reference Books:

1. Pierre D. A., "Optimization, Theory with Application", John Wiley & sons.
2. R. V. Rao, "Decision Making in the Manufacturing Environment Using Graph Theory and Fuzzy Multiple Attribute Decision Making" (Springer Publication).
3. Ritter H., Martinetz T., & Schulten K., Addison, "Neural Computation and Self-Organizing Maps"- Wesley Publishing Company
4. Douglas C. Montgomery, "Design and analysis of experiments" (John Wiley & Sons Inc.)
9. Saravanan R, "Manufacturing Optimization through Intelligent Techniques", Taylor & Francis (CRC Press) - 2006.

Online References:

Sr. No.	Website Name
1	https://nptel.ac.in/courses/112/101/112101298/ - Optimization from Fundamentals, IIT Bombay

Assessment:

Internal Assessment (IA) for 40 marks:

- IA will consist of Two Compulsory Internal Assessment Tests of 20 marks each. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 60% to 50% of syllabus content must be covered in Second IA Test

End Semester Examination for 60 Marks

Question paper format

- Question Paper will comprise of a total of six questions each carrying 15 marks Q.1 will be compulsory and should cover maximum contents of the syllabus
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of Four questions needs to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20351143	Advanced Sensors and Actuators	3	-	-	3	-	-	3

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
20351143	Advanced Sensors and Actuators	20	20	40	60	2	--	--	100

Rationale:

Modern robotics and automation systems require advanced sensors and actuators for intelligent interaction with the environment. This course bridges the gap between macroscopic robotic control and the microscopic principles of sensor and actuator devices. It focuses on energy conversion, material properties, MEMS fabrication techniques, and the miniaturization of sensing and actuation systems.

Course Objectives:

1. To understand the role of sensors, actuators, and energy transformation in robotics and automation.
2. To study characteristics, calibration, reliability, and safety aspects of sensing systems.
3. To gain knowledge of sensor fabrication using thin films and deposition techniques.
4. To understand smart sensors, signal conditioning, and interfacing methods.
5. To learn fundamentals of micro-sensors and MEMS fabrication techniques.
6. To study practical applications of sensors in automation and robotic systems.

Course Outcomes:

1. Apply principles of sensors, actuators, and transducers in robotic systems.
2. Analyze static and dynamic characteristics of sensors and evaluate measurement performance.
3. Analyze thin-film materials and fabrication techniques used in sensor systems.
4. Apply smart sensors and signal conditioning techniques in practical applications.
5. Analyze MEMS-based micro-sensors and determine their suitability for applications.
6. Select and apply appropriate sensors for automation and robotics using real-world case studies

DETAILED SYLLABUS:

Sr No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	To understand this subject, student should have a Basic knowledge of Electrical and Electronics Engineering, Analog and Digital Electronics, and Sensors and Instrumentation is required. Understanding of physics fundamentals (semiconductors, materials, energy principles) along with basic concepts of control systems and robotics is also recommended.	0	
I	Sensors, Actuators and Energy Transformation in Robotics	1.1 Role of sensors and actuators in automation and robotics. 1.2 Transducers and energy conversion principles 1.2 Classification of robotic sensors and actuators 1.3 Static and dynamic characteristics 1.4 Calibration techniques for robotic sensing systems 1.6 Reliability and safety considerations.	4	CO1
II	Sensor Fabrication	2.1 Thin film materials and physical properties 2.2 Thin films in MOSFETs and smart sensor electronics 2.3 Chemical Vapour Deposition techniques 2.3.1APCVD, LPCVD, PECVD, MOCVD 2.4 Physical Vapour Deposition techniques 2.3.1Thermal evaporation, sputtering, PLD 2.5 Fabrication challenges in robotic sensor systems	8	CO2
III	Smart sensors	3.1 Smart sensor basics ,architecture of smart sensors 3.2 Signal conditioning of smart sensors, A to D conversion. 3.3 applications of smart sensors. 3.4 DHT (Digital humidity and temperature sensor) sensors, types, working of the same. 3.5 ADXL345 Accelerometer	8	CO3
IV	Micro Sensors MEMS	4.1 Introduction and characteristics 4.2 Photolithography 4.3 Wet & dry etching (RIE, DRIE) 4.4 MEMS applications	6	CO4
V	Sensors and Advanced Actuators in Robotics	5.1 Gas sensors for environmental monitoring 5.2 Proximity and vision sensors 5.3 Encoders and LiDAR sensors 5.4 Introduction to advanced actuators in robotics 5.5 Piezoelectric actuators 5.6 Shape Memory Alloy (SMA) actuators	8	CO5

		5.7 Electroactive Polymer (EAP) actuators 5.8 Applications of advanced actuators in robotics		
VI	Case Studies	6.1 Vision and imaging sensors 6.2 Proximity and tactile sensors 6.3 Navigation and motion sensors 6.4 Environmental, acoustic & light sensors	5	CO6

Text Books:

1. Madou, M. J., "Fundamentals of Microfabrication and Nanotechnology," 3rd Edition, CRC Press, 2011.
2. Fraden, J., "Handbook of Modern Sensors: Physics, Designs, and Applications," 5th Edition, Springer, 2016.
3. Ananthasuresh, G. K., Vinoy, K. J., Gopalakrishnan, S., Bhat, K. N., & Aatre, V. K., "Micro and Smart Systems," 1st Edition, Wiley India, 2012.
4. Sawhney, A. K., & Sawhney, P., "A Course in Electrical and Electronic Measurements and Instrumentation," Dhanpat Rai & Co., 2017.
5. de Silva, C. W., "Sensors and Actuators: Engineering System Instrumentation," 2nd Edition, CRC Press, 2015.
6. Uchino, K., "Emerging Actuator Technologies: A Micromechatronic Approach," CRC Press, 2006.

References:

1. The absolute standard for Sensor Physics : Fraden, J., "Handbook of Modern Sensors: Physics, Designs, and Applications," 5th Edition, Springer, 2016.
2. The definitive reference for Fabrication and Process Physics: Madou, M. J., "Fundamentals of Microfabrication and Nanotechnology," 3rd Edition, CRC Press, 2011.
3. The best resource for System Integration and Smart Systems : Ananthasuresh, G. K., Vinoy, K. J., Gopalakrishnan, S., Bhat, K. N., & Aatre, V. K., "Micro and Smart Systems," 1st Edition, Wiley India, 2012.

Online References:

Sr. No.	Website Name
1.	https://onlinecourses.nptel.ac.in/noc26_ee41/preview
2.	https://onlinecourses.nptel.ac.in/noc26_ee63/preview

Assessment:

Internal Assessment (IA) for 40 marks:

- IA will consist of Two Compulsory Internal Assessment Tests of 20 marks each. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 40% to 50% of syllabus content must be covered in Second IA Test

End Semester Examination for 60 Marks

Question paper format

- Question Paper will comprise of a total of six questions each carrying 15 marks Q.1 will be compulsory and should cover maximum contents of the syllabus
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of Four questions needs to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2035115	Power Electronics and Drives Lab	-	2	-	-	1	-	1

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
2035115	Power Electronics and Drives Lab	--	--	--	--	--	25	25	50

Course Objectives:

1. To impart knowledge about various power semiconductor devices related to its characteristics, ratings, protection and to select semiconductor devices for various applications.
2. To introduce different methods of power conversion such as ac to dc, dc to dc, dc to ac the underlying principles of converter operation and hence to analyze different converter circuits for power conversion.
3. To keep abreast with the latest technologies and research going on in different domains related to power electronics.
4. To introduce apply power electronics concepts in automation and robotics systems.
5. To develop analytical skills for industrial drive applications.
6. To Acquire and compare the converter circuits such as rectifiers, choppers, and inverters.

Course Outcomes:

1. Draw V-I characteristics of power electronic devices and analyze the firing circuit of SCR
2. Analyze various single phase and three phase power converter circuits and understand their applications.
3. Analyze DC to DC & DC to AC converter circuits and their applications
4. Identify and describe various auxiliary circuits and requirements in power electronics applications such as gate driver circuit, snubber circuits and heat sinks
5. Simulate the performance of Power electronic drive system.
6. Perform various auxiliary circuits and explain the working and applications of electrical drives.

Suggested List of Laboratory Experiment:

Sr. No.	Suggested List of Experiments	LO	Hrs
1	V-I Characteristics of SCR, and UJT triggering Circuit	LO 1	2
2	Study and comparison of IGBT, MOSFET and Power BJT.	LO 1	2
3	Single phase Full-wave controlled Rectifier using SCR with R & RL Loads.	LO 2	2
4	Performance of Single-phase Half wave-controlled rectifier	LO2	2
5	Performance of Single-phase Full-wave controlled rectifiers with R and RL load	LO2	2
6	Performance of 3- phase half wave controlled rectifier	LO3	2
7	Single phase midpoint Cyclo-converter with R load.	LO 3	2
8	Performance of step-up chopper for different values of duty cycles	LO 4	2
9	Step-down chopper for R and RL load	LO 4	2
10	To study speed control of a 3-phase induction motor using V/f control method.	LO 5	2
11	Take any applications of drives in Robotics and Automation system	LO 5	2
12	Light dimmer circuit using DIAC-TRIAC	LO 2	2
13	To study the speed control of a DC motor using a DC drive (armature voltage control method).	LO 5	2
14	To study speed–torque characteristics of motor under different loads.	LO 5	2

Sr. No.	List of Assignments / Tutorials	Hrs
1	Compare and contrast the characteristics and applications of SCR, TRIAC, MOSFET, IGBT and Power BJT.	2
2	Give various types of Commutation methods of SCR and explain each of them briefly	2
3	Design half and full-wave controlled rectifier with R and RL load circuit using SCRs and explain its working with waveforms.	2
4	Give working principle of operation, performance parameters of Three phase.	2
5	Design and Develop battery charging systems	2
6	Design and Develop Electric and Autonomous machine which work with feedback systems for real-time application.	2

Virtual Lab Website Reference:

Sr.No.	Web Site
1	http://vlab.co.in/broad-area-electrical-engineering
2	http://vlab.co.in/broad-area-electronics-and-communications

Assessment:

Term Work: Term Work shall consist of at least 10 Experiments based on the above list. Also, Term work Journal must include at least 2 assignments.

Distribution of Marks for Term Work (25 Marks):

Laboratory Work	15 Marks
Assignments	05 Marks
Attendance	05 Marks

End Semester Practical/Oral Examination:

1. Pair of Internal and External Examiner should conduct practical / viva based on contents
2. Distribution of marks for practical/viva examination shall be as follows:
 - a. Practical performance 15 Marks
 - b. Viva 10 Marks
3. Evaluation of practical examination to be done based on the experiment performed and the output of the experiment during practical examination
4. Students work along with evaluation report to be preserved till the next examination.

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2035116	Robot Kinematics and Dynamics lab	--	02	-	--	01	-	01

		Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
2035116	Robot Kinematics and Dynamics lab	--	--	--	--	--	25	25	50

Course Objectives:

1. To introduce types of mechanisms for various robotic applications.
2. To get knowledge about basic Geometrical and Algebraic approach to solve forward kinematics of serial manipulator
3. To get knowledge about advanced forward kinematics of serial manipulators.
4. To get knowledge about inverse kinematics of various serial manipulators.
5. To get knowledge about design of robot manipulators based on dynamic analysis
6. To develop understanding of industrial robotic systems through virtual modelling and industrial exposure to robotic applications.

Course Outcomes:

1. Apply knowledge to analyze and select appropriate mechanisms for robotic applications
2. Apply and analyze the Coordinate frames, transformations and Forward kinematics of robots
3. Apply and analyze the Inverse kinematics of robots
4. Apply and estimate robot manipulators based on dynamic analysis
5. Apply methods to analyze and measure the mass moment of inertia and balancing of robotic links
6. Apply and analyze fundamentals of industrial automation

List of Experiments: Minimum Ten experiments.

Sr. No.	List of Experiments	LO Mapping	Hrs
1.	Computer program for analysis and synthesis of any mechanism and test it	LO1	2
2.	Determination of holding torque in epicyclic gear train.	LO1	2
3	Design of cams and followers	LO1	2
4.	Forward kinematics of robots using roboanalyzer	LO2	2

5.	Understanding coordinate frames and transformations using roboanalyzer	LO2	2
6	Inverse kinematics of robots using roboanalyzer	LO3	2
7.	Case Study: Kinematics of MTAB Mini Robot	LO3	2
8.	Inverse and Forward dynamics of robots using robo analyzer	LO4	2
9.	Determination of mass moment of inertia and radius of gyration of robotic links.	LO5	2
10.	Experiment on balancing of mass	LO5	2
11.	Virtual Models of Industrial Robots using robo analyzer	L06	2
12.	Industrial visit	LO6	2

Sr. No.	List of Assignments / Tutorials	Hrs
1	Forward Kinematics (DH Convention): Constructing Denavit-Hartenberg (DH) parameter tables and transformation matrices for serial manipulators (e.g., 2R, 3R, PUMA 560) to determine end-effector position.	2
2	Inverse Kinematics (IK): Solving for joint angles (analytical or numerical methods) given a desired end-effector position.	2
3	Differential Kinematics: Computing the Jacobian matrix to relate joint velocities to end-effector velocities.	2
4	Workspace Analysis: Defining and plotting the reachable work envelope of planar and 3D robots.	2
5	Dynamics: Using Euler-Lagrange equations or Newton-Euler methods to determine the torque required to achieve a desired acceleration, considering inertia and gravity.	2
6	Trajectory Planning: Generating smooth paths (e.g., cubic polynomials) in joint space or operational space.	2

Virtual Labs

- <http://ial-coep.vlabs.ac.in/> - Industrial Automation Laboratory, COEP.

Visualization of DH (Denavit-Hartenberg) parameters in Roboanalyzer (*Roboanalyzer is free software developed by IIT Delhi, available on www.roboanalyzer.com)

Assessment:

Term Work: Term Work shall consist of at least 10 Experiments based on the above list. Also, Term work Journal must include at least 2 assignments.

Distribution of Marks for Term Work (25 Marks):

Laboratory Work 15 Marks

Assignments 05 Marks

Attendance 05 Marks

End Semester Practical/Oral Examination:

1. Pair of Internal and External Examiner should conduct practical / viva based on contents

2. Distribution of marks for practical/viva examination shall be as follows:

a. Practical performance 15 Marks

b. Viva 10 Marks

3. Evaluation of practical examination to be done based on the experiment performed and the output of the experiment during practical examination
4. Students work along with evaluation report to be preserved till the next examination

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2035117	Digital Signal Processing Lab	--	02	-	--	01	-	01

Course Code	Course Name	Examination Scheme						
		Theory Marks				Term Work	Practical/ Oral	Total
		Internal assessment			End Sem. Exam			
		Test1	Test 2	Avg. of 2 Tests				
2035117	Digital Signal Processing Lab	--	--	--	--	25	25	50

Course Objectives:

1. To make the students familiarize with discrete-time signals and systems and basic signal operations.
2. To develop the ability to perform frequency-domain analysis using transforms and to interpret system behavior.
3. To enable students to implement DFT and FFT algorithms and use them for efficient spectral analysis and fast convolution.
4. To build competence in designing and analyzing IIR digital filters
5. To develop skills in designing FIR digital filters
6. To expose students to DSP processors and real-time DSP applications

Course Outcomes:

1. Apply concepts of discrete-time signals and LTI systems
2. Analyse signals and systems in the frequency domain
3. Compute and interpret DFT/FFT results
4. Design and simulate IIR digital filters
5. Design, simulate, and evaluate FIR digital filters
6. Apply DSP algorithms and processors to implement real-time or simulated signal processing applications

List of Experiments.

Sr. No	List of Experiments	LO Mapping	Hrs.
01	Write a Program to implement Linear Convolution of the two given sequences.	LO1	02
02	Write a Program to obtain the auto-correlation and Cross-correlations of the given sequences.	LO1	02
03	Compute autocorrelation and cross-correlation; estimate signal energy and power.	LO1	02
04	Write a Program to obtain the circular convolution of the two given sequences.	LO2	02
05	Compute Z-transform of standard sequences, plot poles & zeros and reason about ROC/stability/causality.	LO3	02
06	Write a Program to obtain the linear convolution using circular convolution of two given sequences.	LO3	02
07	Compute linear convolution via direct sum and using circular convolution (DFT method); compare results.	LO3	02
08	Implement radix-2 decimation-in-time and decimation-in-frequency FFT and compare with built-in FFT.	LO3	02
09	Implement fast convolution using FFT; compare complexity and results to direct convolution.	LO4, LO5	02
10	Write a Program to find the DFT of the given sequences. Plot its magnitude and phase plot. Also find its IDFT to obtain the original sequence.	LO4, LO5	02
11	Familiarize with DSP processor architecture and implement simple DSP kernels (FIR/FFT) in fixed-point arithmetic.	LO6	02
12	Write a Program to design low-pass and high-pass IIR filters using window functions.	LO6	02
13	Write a Program to design low-pass and high-pass FIR filters using window functions.	LO6	02
14	Design Butterworth/Chebyshev analog low-pass prototype and transform to digital using bilinear transform	LO6	01

Sr. No.	List of Assignments / Tutorials	Hrs
1	Apply and analyze discrete-time signal operations and convolution for given signals	2
2	Apply Z-transform and DTFT to analyze properties, stability, and causality of discrete-time systems	2
3	Apply and analyze DFT and FFT algorithms for efficient computation of signal spectra.	2
4	Apply and analyze IIR filter design techniques using analog-to-digital transformation methods	2

5	Apply and analyze FIR filter design using various windowing techniques	2
6	Apply and analyze DSP processors for implementing real-world signal processing applications.	2

Assessment:

Term Work: Term Work shall consist of at least 10 Experiments based on the above list. Also, Term work Journal must include at least 2 assignments.

Distribution of Marks for Term Work (25 Marks):

Laboratory Work 15 Marks

Assignments 05 Marks

Attendance 05 Marks

End Semester Practical/Oral Examination:

1. Pair of Internal and External Examiner should conduct practical / viva based on contents
2. Distribution of marks for practical/viva examination shall be as follows:
 - a. Practical performance 15 Marks
 - b. Viva 10 Marks
3. Evaluation of practical examination to be done based on the experiment performed and the output of the experiment during practical examination
4. Students work along with evaluation report to be preserved till the next examination

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20351181	Data Structures and Algorithms	2	-	-	2	-	-	2

Course Code	Course Name	Examination Scheme						
		Theory Marks				Term Work	Practical/ Oral	Total
		Internal assessment			End Sem. Exam			
		Test 1	Test 2	Avg. of 2 Tests				
20351181	Data Structures and Algorithms Lab	--	--	--	--	25	25	50

Course Objectives:

1. To learn mathematical background for analysis of algorithm
2. To learn various advanced data structures.
3. To learn greedy approach to solve problems.
4. To learn and use dynamic programming algorithms
5. To learn and use backtracking and branch and bound approach.
6. To learn and use pattern matching algorithms

Course Outcomes:

1. Understand methods for analysis of algorithms and solve recurrence problems.
2. Choose appropriate advanced data structures for a given problem and calculate its complexity.
3. Analyze the greedy programming technique to solve problems.
4. Analyze the dynamic programming technique to solve problems.
5. Evaluate and analyze the backtracking algorithm and branch and bound algorithms
6. Select a proper pattern matching algorithm for a given problem.

List of Experiments.

Sr No	List of Experiments	LO Mapping	Hrs
01	Experiment based on common mathematical functions. (a. Selection sort, b. Insertion sort)	LO 1	02
02	Experiment based on divide and conquers approach. (a. Merge sort, b. Quick sort, c. Binary Search)	LO1	02
03	Experiment based on divide and conquers approach: Min Max Algorithm	LO1	02
04	Experiment based on divide and conquers approach: Strassen's Algorithm	LO2	02

05	Experiment based on greedy approach. (Single source shortest path- a. Dijkstra b. Fractional Knapsack problem, Minimum cost spanning trees- c. Kruskal and d. Prim's algorithm)	LO2	02
06	Experiment based on greedy approach: Implementation of Job Sequencing with deadlines.	LO 3	02
07	Experiment using dynamic programming approach (All pair shortest path- a. Floyd Warshall, b. 0/1 knapsack)	LO 3	02
08	Experiment using dynamic programming: a. Travelling salesperson problem b. Longest common subsequence	LO 4	02
09	Experiment using dynamic programming: Bellman Ford Algorithm	LO 4	02
10	Experiment using Backtracking strategy. (a. N-queen problem, b. Sum of subsets, c. Graph coloring)	LO 5	02
11	Experiment using branch and bound strategy.	LO 5	02
12	Experiment based on string matching/amortized analysis (a. The Naïve string- matching Algorithms, b. The Rabin Karp algorithm, c. The Knuth-Morris-Pratt algorithm.	LO 6	02

Sr No	List of Assignments / Tutorials	Hrs
01	Assignment covers the topics from Module 1	2
02	Assignment covers the topics from Module 2	2
03	Assignment covers the topics from Module 3	2
04	Assignment covers the topics from Module 4	2
05	Assignment covers the topics from Module 5	2
06	Assignment covers the topics from Module 6	2

Assessment:

Term Work: Term Work shall consist of at least 10 Experiments based on the above list. Also, Term work Journal must include at least 2 assignments.

Distribution of Marks for Term Work (25 Marks):

Laboratory Work	15 Marks
Assignments	05 Marks
Attendance	05 Marks

End Semester Practical/Oral Examination:

1. Pair of Internal and External Examiner should conduct practical / viva based on contents
2. Distribution of marks for practical/viva examination shall be as follows:
 - a. Practical performance 15 Marks
 - b. Viva 10 Marks
3. Evaluation of practical examination to be done based on the experiment performed and the output of the experiment during practical examination
4. Students work along with evaluation report to be preserved till the next examination

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20351182	Optimization Techniques Lab	--	02	-	--	01	-	01

Course Code	Course Name	Examination Scheme						
		Theory Marks				Term Work	Practical/ Oral	Total
		Internal assessment			End Sem. Exam			
		Test1	Test 2	Avg. of 2 Tests				
20351182	Optimization Techniques Lab	--	--	--	--	25	25	50

Course Objectives:

1. Apply practical design problems having two design variables, solve them graphically and analyze the nature of the problem.
2. Apply the simplex method algorithm and solve LPP by two-phase simplex method numerically.
3. Apply algorithm of simplex method to solve quadratic programming problem numerically.
4. Analyze the necessary and sufficient conditions and Estimate the descent conditions for a given optimization problem.
5. Estimate step size along search direction using numerically search methods
6. Apply numerical methods algorithms to solve unconstrained problems.

Course Outcomes:

- 1 Apply practical design problems with two design variables and solve them graphically to identify and analyze the nature of the optimization problem.
2. Apply the simplex method algorithm and solve Linear Programming Problems (LPP) using the two-phase simplex method, and analyze the results numerically.
3. Apply the simplex method to solve quadratic programming problems numerically and analyze the results using simulation software.
- 4 .Estimate the necessary and sufficient conditions for unconstrained optimization problems, and analyze the descent conditions for a given search direction.
- 5 Apply search methods to estimate the step size along a search direction and analyze the results numerically using the Equal Interval Search method.
6. Apply numerical optimization methods such as Newton's method to solve nonlinear optimization problems and analyze the results using simulation software.

List of Experiments.

Sr No	List of Experiments	LO Mapping	Hrs
01	Formulate engineering system design problem as an optimization problem.	LO1	02
02	Problem formulated in Experiment No. 1 should be solved graphically and identify the nature of problem.	LO1	02
03	By using excel solver solve unconstrained and constrained optimization problems create excel worksheets.	LO2	02
04	Solve LPP by two-phase simplex method numerically and verify the results by using simulation software	LO3	02
05	Solve quadratic programming problem numerically and verify results by using simulation software.	LO4	02
06	Verify the descent conditions for a given search direction for unconstrained optimization problem and calculate step size along search direction using Equal Interval Search method numerically and verify results by using simulation software	LO5	02
07	Verify the descent conditions for a given search direction for unconstrained optimization problem and calculate step size along search direction using Golden Section Search method numerically and verify results by using simulation software	LO5	02
08	Solve nonlinear optimization problems by using numerical optimization methods (indirect) steepest-descent and conjugate-gradient methods verify the results by using simulation software.	LO6	02
09	Solve nonlinear optimization problems by using numerical optimization methods (indirect) Newtons methods verify the results by using simulation software.	LO6	02
10	Solve nonlinear optimization problems by using numerical optimization methods (indirect) DFP and BFGS methods verify the results by using simulation software.	LO6	02

Sr No	List of Assignments / Tutorials	Hrs
01	Assignment covers the topics from Module 1	2
02	Assignment covers the topics from Module 2	2
03	Assignment covers the topics from Module 3	2
04	Assignment covers the topics from Module 4	2
05	Assignment covers the topics from Module 5	2
06	Assignment covers the topics from Module 6	2

Assessment:

Term Work: Term Work shall consist of at least 10 Experiments based on the above list. Also, Term work Journal must include at least 2 assignments.

Distribution of Marks for Term Work (25 Marks):

Laboratory Work 15 Marks

Assignments 05 Marks

Attendance 05 Marks

End Semester Practical/Oral Examination:

1. Pair of Internal and External Examiner should conduct practical / viva based on contents
2. Distribution of marks for practical/viva examination shall be as follows:
 - a. Practical performance 15 Marks
 - b. Viva 10 Marks
3. Evaluation of practical examination to be done based on the experiment performed and the output of the experiment during practical examination
4. Students work along with evaluation report to be preserved till the next examination.

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20351183	Advanced Sensors and Actuators Lab	-	2	-	-	1	-	1

Course Code	Course Name	Examination Scheme						
		Theory Marks				Term Work	Practical/ Oral	Total
		Internal assessment			End Sem. Exam			
		Test1	Test 2	Avg. of 2 Tests				
20351183	Advanced Sensors and Actuators Lab	--	--	--	--	25	25	50

Course Objectives:

1. To understand robotic sensors, actuators and transduction mechanisms.
2. To study thin-film materials and sensor fabrication techniques.
3. To interface and analyze smart sensors used in robotics.
4. To understand MEMS fabrication and etching processes.
5. To demonstrate robotic sensors and advanced actuators.
6. To analyze practical applications of robotic sensing systems

Course Outcomes:

1. Apply sensor and actuator principles in laboratory experiments.
2. Analyze sensor fabrication and thin-film technologies.
3. Apply smart sensors in robotic systems.
4. Analyze MEMS fabrication processes and applications.
5. Apply advanced actuators in robotic applications.
6. Analyze robotic sensing systems used in automation.

List of Experiments: Minimum Ten experiments to be performed

Sr No	List of Experiments	LO mapping	Hrs
01	Identification and functional study of common robotic sensors and actuators.	LO1	2
02	Study of transduction mechanisms (electrical, mechanical, thermal).	LO1	2
03	Calibration of sensors (temperature, pressure, force, or displacement).	LO1	2
04	Study and comparison of thin-film materials used in MEMS sensors.	LO2	2
05	Simulation or demonstration of CVD and PVD techniques (APCVD, LPCVD, sputtering).	LO2	2
06	Analysis of thin-film layers in MOSFET-based sensor structures.	LO2	2
07	Design of photomasks and study of photolithography steps.	LO4	2

08	Comparative study of wet etching and dry etching.	LO4	2
09	Study of Smart sensors with any one application.	LO3	2
10	Mini project based on DHT sensor.	LO3	2
11	Study of gas, proximity and vision sensors used in robotics.	LO5	2
12	Interfacing of encoder and LiDAR sensors for robotic applications.	LO5	2
13	Demonstration and analysis of Piezoelectric and SMA actuators.	LO5	2
14	Case study on navigation, tactile and environmental sensors in robotic systems.	LO6	2

Sr No	List of Assignments / Tutorials	Hrs
01	Study and classification of robotic sensors, actuators and transducers used in automation systems.	2
02	Comparative analysis of CVD and PVD techniques used in sensor fabrication.	2
03	Report on architecture and applications of smart sensors in robotics.	2
04	Study of MEMS fabrication processes and applications in robotic systems.	2
05	Comparative study of Piezoelectric, SMA and EAP actuators used in robotics.	2
06	Case study on robotic sensing systems used in navigation, imaging and environmental monitoring.	2

Assessment:

Term Work: Term Work shall consist of at least 10 Experiments based on the above list. Also, Term work Journal must include at least 2 assignments.

Distribution of Marks for Term Work (25 Marks):

Laboratory Work	15 Marks
Assignments	05 Marks
Attendance	05 Marks

End Semester Practical/Oral Examination:

1. Pair of Internal and External Examiner should conduct practical / viva based on contents
2. Distribution of marks for practical/viva examination shall be as follows:
 - a. Practical performance 15 Marks
 - b. Viva 10 Marks
3. Evaluation of practical examination to be done based on the experiment performed and the output of the experiment during practical examination
4. Students work along with evaluation report to be preserved till the next examination

Vertical – 5

Indian Knowledge System-II

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2035511	Indian Knowledge System-II	--	2*+2	-	--	2	--	2

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Avg.					
2035511	Indian Knowledge System-II	--	--	--	--	--	50	--	50

Rationale:

India has ancient Indian knowledge systems (IKS) that have significantly contributed to mathematics, science and engineering etc. NEP 2020 visualizes a framework where students can apply traditional knowledge with contemporary approaches to suggest sustainable solutions and innovative practices. This subject aims to integrate the IKS into present-day engineering education, research, and practice to cultivate well-rounded individuals grounded in India's great intellectual traditions

Course Objectives:

To impart necessary knowledge and skills so as to enable students to

1. Understand the premise of Indian Knowledge System (IKS) in the domain of town planning and governance
2. Appreciate the ancient technologies used in Irrigation planning and transport systems, astronomy and Vastu shastra (The science of architecture) for fulfilling basic human needs.
3. To introduce students to traditional Indian chemical engineering knowledge and chemical processes related to metallurgy, pharmaceuticals.
4. To understand traditional chemical engineering practices, rasashastra apparatus, and to compare traditional processes with modern chemical engineering processes.
5. To enable students to understand India's rich Indigenous Knowledge Systems (IKS) in metals and metal processing.
6. To introduce the early development of transmission systems in India.

Course Outcomes:

After completion of this course, learner will be able to

1. Compare the ancient knowledge in the domain of town planning, governance, and traditional principles in residential (Vastu) construction.
2. Evaluate the developments in irrigations systems, transport systems and astronomy.
3. Explain traditional Indian chemical engineering knowledge and their scientific principles.

4. Describe classical chemical processes, rasashastra apparatus, and indigenous material processing techniques and compare traditional chemical engineering processes with modern chemical engineering processes.
5. Evaluate India's indigenous metal and metal-processing heritage.
6. Analyze early Indian transmission technologies

Prerequisite: Basic concepts of Science and Technology

Detailed Syllabus:

Module No.	Unit no	Detailed Content	Hours	CO Mapping
1	Title: Town Planning and Governance, Vastu shastra – The science of Architecture		5	CO1
	1.1	Development of ancient text, principles, Terminologies, Indic languages, various typologies, social infrastructure-places for gathering, worship, entertainment, cremation etc. Perspective of Arthaśāstra on town planning.		
	1.2	Governance systems and Principles of Town Planning: Principles of land management, local and global trade routes and trade centres; the establishment and decay of urban settlements. Ancient India urbanization, layouts in terms of geometry, population, geography, and various communities. Evolution of the cities, planning principles in urban centres. interconnection between ancient knowledge of town planning and present planning practices.		
	1.3	Eight limbs of Vastu, Ancient literature and archaeological evidence of building and building materials, Fortification and Forts.		
	1.4	Structural Principles in worship places Architecture. marvelous stone architecture for eternity, Iconography Use of bricks or dressed stone based on logical principals of structural engineering.		
2	Irrigation planning, transport systems and Astronomy		4	CO2
	2.1	Irrigation and water management. Ancient irrigation systems, dams, reservoirs, sluices, canals, embankments, wells, water-lifting devices, etc. from a historical point of view. Irrigation systems and practices in South India		
	2.2	Ancient transport systems during various periods: Indus valley, Buddhist period, Mauryan period, Gupta period, etc. Shipbuilding: Indigenous shipbuilding, materials and methods-Use of local materials: teak, bamboo, coir, and lac-		

		Construction techniques: -Innovations in hull design, masts, and rigging.		
	2.3	Indian Astronomy: Planetary System. Motion of the Planets; Velocity of Light; Eclipse. Astronomy. Navagrahas. Important works in Indian Astronomy. Aryabhata and Nilakantha: Contribution to Astronomical Studies		
3	Traditional Chemical Processes and Industrial Advancements		4	CO3
	3.1	Overview of ancient Indian scientific texts related to chemicals, Arthashastra – mining techniques and metallurgy,		
	3.2	Charaka Samhita – pharmaceutical preparations and fermentation, Sushruta Samhita – preparation of alkali and mineral medicines, traditional chemical operations and processes such as manual metal extraction methods and herbal extraction techniques, small-scale batch production methods,		
	3.3	production of natural alkali from plant ash, comparison of traditional metallurgy with modern metallurgical processes, herbal extraction with solvent extraction and chromatography, natural alkali production with chlor-alkali industry, batch processing with continuous processing.		
4	Rasashastra – Ancient Indian Chemical Science and Classical Apparatus		4	CO4
	4.1	Introduction, scope of rasashastra in traditional Indian alchemy and medicine, development of rasashastra, contributions of Nagarjuna in metallurgical and alchemical practices, role of mercury (Parada) and metals in classical medicinal formulations,		
	4.2	Various apparatus used in rasashastra such as Dolayantra (boiling and heating setup), Patana Yantra (distillation apparatus), Puta (system for controlled heating), Musha (crucible) used for melting and calcination, Khalva Yantra (grinding and mixing of materials), rasashastra processes such as Shodhana (purification), Marana (calcination), Jarana (roasting) and Bhavana (levigation).		
	4.3	Comparison of Dolayantra with modern jacketed reactors, Patana Yantra with distillation columns, Marana process with rotary kiln or muffle furnace, mercury purification methods with modern hydrometallurgical refining techniques.		
5	Indigenous Metal and Metal Processing Techniques		5	CO5
	5.1	Early Metals in India Use of copper, bronze, gold, silver, tin, and lead in Harappan and Vedic periods. Indigenous mining, ore beneficiation, furnace designs, and smelting practices.		

	5.2	Landmark Indian Metallurgical Achievements Wootz Steel: crucible steelmaking, carbon control, carbide banding, global influence as Damascus steel. Zawar Zinc Distillation: world's earliest large-scale zinc production; downward distillation furnaces. Chola Bronze Metallurgy: alloy design (Cu–Sn–Pb), precision lost-wax casting for sculptures. Aranmula Metal Mirror: high-tin bronze polishing technology with mirror-grade reflectivity. Konark Temple Iron Beams & Clamps: structural iron demonstrating marine corrosion resistance.		
	5.3	Scientific Insights Behind Traditional Metallurgy Delhi Iron Pillar: natural formation of a passive protective layer; high-phosphorus wrought iron. Microstructure evolution through traditional forming and heat-treatment cycles. Traditional Indian blacksmiths (Lohar, Vishwakarma, Asari, Kammalar communities)		
6		Development of Transmission in India	4	CO6
	6.1	Transmission in Early Indian Civilizations Indus Valley Civilization: evidence of wheel technology, carts, pottery wheels, and early bearing surfaces. Use of pulleys, levers, and inclined planes in construction and material handling.		
	6.2	Traditional gear and motion Conversion systems Sugarcane crushers, oil ghanis, temple clocks, and granaries		

Textbooks

1. Pride of India: A Glimpse into India's Scientific Heritage, Samskrita Bharati, New Delhi.
2. Mahadevan, B., Bhat Vinayak Rajat, Nagendra Pavana R.N. (2022), "Introduction to Indian Knowledge System: Concepts and Applications", PHI Learning Private Ltd. Delhi.
3. Kak, S.C. (1987). "On Astronomy in Ancient India", Indian Journal of History of Science, 22(3), pp. 205–221.
4. Bag, A.K. (1997). History of Technology in India, Vol. I, Indian National Science Academy, New Delhi.
5. Acarya, P.K. (1996). Indian Architecture, Munshiram Manoharlal Publishers, New Delhi.
6. Banerjea, P. (1916). Public Administration in Ancient India, Macmillan, London.
7. Gagan Bansal, Indian Knowledge System: Integrating Heritage with Engineering
8. Prafulla Chandra Ray, A History of Hindu Chemistry –
9. R. Balasubramaniam – Wootz Steel: An Advanced Material of the Ancient World
10. K.T.M. Hegde – Indian Archaeometallurgy
11. T.R. Anantharaman – The Rustless Wonder: The Iron Pillar of Delhi
12. AICTE – IKS Division Publications

References

1. Kapoor Kapil, Singh Avadhesh (2021). "Indian Knowledge Systems Vol – I & II", Indian Institute of Advanced Study, Shimla, H.P
2. Subbarayappa, B.V. and Sarma, K.V. (1985). Indian Astronomy: A Source Book, Nehru Centre, Mumbai.
3. R.T.V. Ramachandra – History of Indian Metallurgy
4. A. Ghosh - Encyclopedia of Indian Archaeology
5. Research papers on Zawar zinc distillation, Chola bronzes, and Aranmula mirror metallurgy.

Online References:

Sr. No.	Website Name
1.	https://onlinecourses.swayam2.ac.in/e-learning/preview/imb23_mg54
2.	https://iksindia.org
3.	https://iks.iitk.ac.in

The experiential learning sessions may include:

1. Field Visits to historical sites, museums.
2. Arranging students' discussions with experts in various fields of IKS.
3. Providing the students with online lectures by distinguished experts in the field.
4. Students' participation in activities related to understand the practical aspects of Indian knowledge systems.
5. Traditional practices of arts, crafts, agriculture, Vastu Shastra etc.,
6. Herbal extraction using decoction or distillation methods.
7. Preparation of lime mortar (calcination, slaking, carbonation).
8. Natural dye extraction and mordanting process.
9. Herbal soap preparation through saponification.
10. Comparative study of batch vs continuous processing.

Sr No	Details of Activities
01	Project-based activities
02	Presentation
03	Case studies
04	Visit Museum/historical places and write a report
05	Flip class mode
06	Quiz with MCQ
07	Comparative Study of IKS & other scientific systems around the world
08	Group Discussions
09	Roleplay
10	Self-study activities

Note: The faculty can choose any of these activities for continuous assessment

Assessment:

Term work Assessment for 50 marks:

Suggested Pedagogy and assessment criteria:

1. Assignment as per the modules. 06
2. Project-based activities.
3. Presentation, Group Discussions, and Case studies.

4. Visit historical places.
5. Flip class mode/ Roleplay
6. Quiz MCQ
7. Assessment through flipped class and PowerPoint presentation along with documentation

Term Work:

Assignments	10 Marks
Presentation	20 Marks
Quiz /MCQ	15 Marks
Attendance	05 Marks

Semester – VI

Vertical – I

Major

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2036111	Robotic Control System	3	-	-	3	-	-	3

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
2036111	Robotic Control System	20	20	40	60	2	--	--	100

Rationale:

The course Robotic Control Systems forms a crucial link between the mechanical design and intelligent operation of robots. Modern automation and robotics industries demand engineers who not only understand the kinematics and dynamics of manipulators but can also design and implement control strategies that ensure accuracy, stability, and adaptability. This subject introduces learners to both classical and modern control techniques used in robotic systems, integrating mechanical, electrical, and computational principles. It strengthens analytical and problem-solving skills through modeling, simulation, and real-time implementation of control algorithms.

Course Objectives:

1. To develop the ability to model the dynamics of robotic manipulators.
2. To enable students to analyze nonlinear control strategies for robotic manipulators.
3. To provide insight to apply and evaluate force control methods in robotic systems.
4. To develop competence to model and interpret the kinematics of steered mobile robots. To enable students to apply and explore vision-based control approaches in robotics.
5. To enable students to apply and evaluate classical and resolved motion rate control techniques.
6. Implement and evaluate control algorithms for robotic systems through case studies such as PID control of manipulators, resolved motion rate control, and industrial robot control applications.

Course Outcomes:

1. Apply the basic concepts in dynamic modeling of robotic manipulators.
2. Analyze and design nonlinear control schemes for robotic manipulators.
3. Design and implement force control strategies for manipulators.
4. Estimate and analyze the kinematic models of steered and differentially driven mobile robots.
5. Apply vision-based techniques for robotic control systems.
6. Design and evaluate PID control strategies for single-link and planar 2R manipulators.

DETAILED SYLLABUS:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Concepts of open-loop and closed-loop systems, Transfer functions, and stability analysis, Laplace transforms, matrix algebra, and differential equations, Basics of forward and inverse kinematics, degrees of freedom, and dynamic modeling using Newton–Euler or Lagrange’s methods, understanding of sensors, actuators, and DC/AC motor control. (Note: There will be no questions from the prerequisite in the theory examination)	00	
I	Review of dynamic modelling of robots:	Typical block schematic of robotic control systems and components, Review of dynamic modelling of robots using Newton–Euler and Lagrangian formulations, Linear control of manipulators -second-order system representation and stability basics, Concept of control-law partitioning and trajectory following control, Feedback control examples: single-link manipulator and wall-following mobile robot. #Self-learning topics: Necessity of controllers for robots, Open-loop, and closed-loop control concepts.	08	CO1
II	Nonlinear Control of manipulators	PD with gravity compensation – concept and control law formulation, computed torque control – inverse dynamics approach for trajectory tracking, Basics of adaptive control – handling parameter variations in manipulators, Task-space control schemes – resolved motion rate and acceleration control. #Self-learning topics: Need for nonlinear control in robotic manipulators, limitations of linear control, Basics of adaptive control Overview	07	CO2
III	Force control of manipulators	Force control using a mass–spring system, the hybrid position/force control problem, the hybrid position/force control scheme. Impedance and admittance control concepts. Comparison between position control and force control. #Self-learning topics: Introduction and need for force control in robotic manipulators, Applications of robots in assembly and contact tasks (Note: No questions should be asked on self-learning topics)	06	CO3
IV	Kinematic model of steered mobile robots and differentially driven mobile robots	Control of a mobile robot to move to a point, Control to move the robot to a desired pose, Dynamic model of a quadcopter, Controller design for quadcopter trajectory tracking. #Self-learning topics: Line-following and path-following control strategies (Note: No questions should be asked on self-learning topics)	07	CO4

V	Vision based Control	Image segmentation and interpretation for object detection and tracking, Pose estimation techniques for robotic applications, Camera calibration, and coordinate transformations, Position-based visual servoing (PBVS) – principle and control strategy, Image-based visual servoing (IBVS) – concept and implementation, Hybrid visual servoing – combination of PBVS and IBVS approaches. #Self-learning topics: Configuration and components of a vision system in robotics, Stereo vision and depth perception (Note: No questions should be asked on self-learning topics)	08	CO5
VI	Case study	PID control of a single-link robotic manipulator, PID control of a planar 2R manipulator, Resolved motion rate control of a 2R manipulator, Control implementation of any industrial robot. #Self-learning topics: Force control of peg-in-hole assembly task (Note: No questions should be asked on self-learning topics)	03	CO6

(Note: No questions should be asked on self-learning topics)

Text Books:

1. Robotics and Control, R K Mittal, I J Nagrath
2. Nonlinear systems, Hassan K. Khalil, Pearson
3. Introduction to Robotics Mechanics and Control, John J. Craig, 3e, Pearson.
4. Robotics: Fundamental Concepts and Analysis, Ashitava Ghosal, Oxford
5. Robotics- Modelling planning and control- Bruno Siciliano, Lorenzo Sciavicco Luigi Villani, Giuseppe Oriolo, Springer-Verlag London.
6. Peter Corke, “Robotics, Vision and Control-Fundamental Algorithms in MATLAB”, Springer Tracts in Advanced Robotics, volume 73.
7. The Robotics Primer-Maja J Matarić, The MIT Press

References:

1. Probabilistic Robotics: Sebastian Thrun, Wolfram Burgard, Dieter Fox, MIT Press
2. Modern Robotics Mechanics, Planning and Control, Kevin M. Lynch, Frank C. Park, Cambridge University Press, 2017.

Online References:

Sr. No.	Website Name
1.	NPTEL- Introduction to Robotics: https://nptel.ac.in/courses/112105249
2.	NPTEL- Robotics and Control: Theory and Practice: https://onlinecourses.nptel.ac.in/noc20_me03/preview?utm_source=chatgpt.com
3.	NPTEL- Intelligent Control of Robotic Systems: https://onlinecourses.nptel.ac.in/noc25_ee89/preview?utm_source=chatgpt.com

Assessment:

Internal Assessment (IA) for 40 marks:

- IA will consist of Two Compulsory Internal Assessment Tests of 20 marks each. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 60% to 50% of syllabus content must be covered in Second IA Test

End Semester Examination for 60 Marks

Question paper format

- Question Paper will comprise of a total of six questions each carrying 15 marks Q.1 will be compulsory and should cover maximum contents of the syllabus
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of Four questions needs to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2036112	Process Instrumentation and Control	3	-	-	3	-	-	3

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
2036112	Process Instrumentation and Control	20	20	40	60	2	--	--	100

Rationale:

At the end of this course, the students will be equipped with the essential knowledge and skills for designing, analyzing, and implementing control systems in industrial processes. They will also be able to analyze dynamic behaviour and design single-loop PID-based control strategies, including appropriate actuators and control schemes, for given industrial process case studies, and evaluate expected closed-loop performance.

Course Objectives:

1. To understand the dynamic behaviour of industrial processes and the need for effective control.
2. To explain the purpose and performance impact of different industrial controllers
3. To identify and apply major control strategies such as feedback, feed-forward, cascade, and adaptive control.
4. To understand the working and role of pneumatic, hydraulic, and electromechanical actuators.
5. To recognize key industrial operations and equipment along with their control requirements
6. To relate theoretical concepts to real industrial applications through case studies

Course Outcomes:

1. Analyze process control terminology, control loop structures, and process dynamics including dead time and inverse response.
2. Evaluate ON-OFF and PID-based controllers using operational equations and transfer functions.
3. Apply and construct block-diagram models of feedback, feedforward, cascade, ratio, split-range, selective, adaptive, and inferential control schemes.

4. Differentiate and assess pneumatic, hydraulic, and electromechanical actuators, including DC, AC, and stepper motors for control applications.
5. Analyze the operation and control requirements of industrial units such as heat exchangers, boilers, and dryers.
6. Analyze industrial applications and justify optimal control strategies for improved performance.

DETAILED SYLLABUS:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Basic components of control system and different types of sensors. Fundamentals of AC and DC, OPAMPs		
I	Introduction to Process Control	1.1 Process Control Terminology, 1.2 Development of Typical Process Control loops like Pressure, Temperature, flow & Level. 1.3 Process characteristics, control system parameters, 1.4 Dynamic elements in a control loop 1.5 Dead time processes and smith predictor compensator. Inverse response behavior of processes and compensator. 1.6 Dynamic behavior of first and second order systems. Interacting and non-interacting systems. 1.7 Development of Mathematical Model for first & second order system with Example.	06	CO1
II	Process control actions and Controllers	2.1 Discontinuous control actions – two position or ON-OFF: Operation, neutral zone 2.2 Continuous control actions – proportional, integral and derivative: operation, output equations, corresponding transfer function and Response graph. 2.3 Composite controllers – PI, PD, PID controllers: operation, output equations, Response graph, comparison, application Electronic op-amp based PI, PD, PID controllers: circuit diagram, equations.	08	CO2
III	Control Schemes	Feedback, Feed forward, cascade, Ratio, split range, selective control, adaptive control, inferential control-Block diagram and examples	08	CO3
IV	Process Transmitters and	4.1 Concept of signal transmission 4.2 2-wire and 4-wire transmitters 4.3 Smart/HART transmitters	06	CO4

	Final Control Elements	4.4 Control valves, types and characteristics. 4.5 I/P and P/I converters 4.6 DC servo motor – characteristics, difference from a normal DC motor 4.7 AC servo motor – characteristics, difference from a normal 2 phase induction motor 4.8 Stepper motor (PM and variable reluctance type): Working and applications, comparison with DC servo motor		
V	Introduction to unit operations and Utilities	5.1 Working principle and list of types of Heat exchanger, Boilers and Dryers. 5.2 Working of Shell and Tube Heat Exchanger and different types of controls associated with it 5.3 Working of Water tube boiler and different types of controls associated with it 5.4 Working of Fluidised bed and Double drum dryer and different types of controls associated with it	07	CO5
VI	Case studies	Case Study 1 On–Off Controller in a Domestic Water Heater- Concepts: Hysteresis/neutral zone, thermostatic behavior, temperature cycling. Analysis: Why PI/PID is not used, effect on comfort and safety Case Study 2 Cascade Control in a Heat Exchanger • Master: Outlet temperature • Slave: Steam flow • Concepts: Cascade advantage, disturbance handling. Discussion: Improvement over single-loop feedback Case Study 3: Servo Motor Position Control in a Robotic Arm • Concepts: DC servo characteristics, closed-loop feedback, stability. • Discussion: Difference from normal DC motors and applications. Case Study 4: Stepper Motor Control in CNC Machines • Plant: Stepper motor + driver • Concepts: Open-loop control, micro stepping, torque-speed characteristics. • DC servo motor: closed-loop, good for variable load and high precision. • Stepper: simpler, but can lose steps under high load if open-loop.	04	CO6

Text Books:

1. Curtis D. Johnson, “Process Control Instrumentation Technology”, PHI /Pearson Education 2002.
2. George Stephanopoulos, “Chemical process control”, PHI-1999.
3. B.L Thereja, Fundamentals of Electrical and Electronics, S. Chand

Reference Books:

1. Bela G. Liptak, "Instrument Engineer's Hand Book – Process Control", Chilton Company, 3rd Edition, 1995.
2. M.Chidambaram, "Computer Control of Processes", Narosa, 2002.
3. Deshpande P.B and Ash R.H, "Elements of Process Control Applications", ISA Press, New York, 1995.
4. D. Patranabis, "Principles of Process Control", Second edition, TMH.
5. F.G. Shinsky, "Process Control System", TMH.
6. Andrew Parr, *Hydraulics and Pneumatics- A technician's and engineer's guide*, Jaico Publishing House, Mumbai
7. W.G Andrews and Williams, *Applied Instrumentation in the process Industries*, Vol. - I and II, Gulf Publication

Online References:

Sr. No.	Website Name
1	https://learncheme.com/
2	https://controlguru.com/
3	https://instrumentationtools.com/
4	https://www.electronics-tutorials.ws/

Assessment:**Internal Assessment (IA) for 40 marks:**

- IA will consist of Two Compulsory Internal Assessment Tests of 20 marks each. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 40% to 50% of syllabus content must be covered in Second IA Test

End Semester Examination for 60 Marks**Question paper format**

- Question Paper will comprise of a total of six questions each carrying 15 marks Q.1 will be compulsory and should cover maximum contents of the syllabus
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of Four questions needs to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20361131	Database Management System (DBMS)	3	1	-	3	1	-	4

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
20361131	Database Management System (DBMS)	20	20	40	60	2	--	--	100

Rationale :

Explore and understand concept of database management system with technologies.

Course Objectives:

1. To understand the fundamentals and architecture of database systems.
2. To design data models using ER/EER techniques and map them to relational models.
3. To apply SQL and advanced SQL for data definition, manipulation, and control.
4. To normalize databases to minimize redundancy and maintain consistency.
5. To understand transaction, concurrency, and recovery techniques in multi-user environments.
6. To explore advanced database technologies such as NoSQL, Cloud, and Graph Databases.

Course Outcomes:

1. Design normalized and optimized database schemas for real-world applications.
2. Implement advanced SQL queries, joins, and stored procedures.
3. Apply indexing and query optimization techniques for performance improvement.
4. Demonstrate understanding of NoSQL and Cloud databases through case studies.
5. Apply and analyze secure database access and transaction control mechanisms.
6. Analyze and implement small-scale database systems integrating SQL and NoSQL.

DETAILED SYLLABUS:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
I	Introduction to Databases & Data Modeling	Overview of DBMS, ACID Properties, 3-Tier Architecture, ER/EER Diagrams, Keys, Relationships, Specialization and Generalization. Self-learning: Case study on ER modeling for e-commerce systems.	6	CO1
II	Relational Model and Advanced Relational Algebra	Relational Model concepts, Schema, Constraints, Mapping from EER to Relational Model, Advanced Operators – Join types, Division, Aggregation, Window Functions. Self-learning: Real-world relational schema design.	6	CO2
III	SQL and Advanced SQL	DDL, DML, DCL, TCL, Integrity Constraints, Nested Queries, Joins, Views, Triggers, Stored Procedures, Common Table Expressions (CTE), Recursive Queries, and Window Functions. Self-learning: Leet Code or Hacker Rank SQL challenges.	8	CO3
IV	Database Normalization & Design	Functional Dependencies, Normal Forms (1NF to BCNF), Decomposition, Denormalization, Data Integrity, Schema refinement. Self-learning: Normalize an existing college database up to 3NF.	6	CO4
V	Transaction Management, Concurrency, and Recovery	Transactions, Serializability, Lock-based and Timestamp-based Protocols, Deadlock Handling, Recovery techniques, Checkpointing, Log-based recovery. Self-learning: Simulate concurrency control using SQL.	7	CO5
VI	Advanced & Modern Databases	Cloud Databases (AWS RDS, Google BigQuery), Distributed DBMS, NoSQL Databases (MongoDB, Cassandra, Redis), Graph Databases (Neo4j), Database Security (Encryption, SQL Injection Prevention), Query Optimization. Self-learning: Implement mini-project integrating SQL and NoSQL databases.	6	CO6

Text Books:

1. Elmasri & Navathe, Fundamentals of Database Systems, Pearson Education, 7th Edition.
2. Raghu Ramakrishnan & Johannes Gehrke, Database Management Systems, TMH.
3. Abraham Silberschatz, Henry Korth, Sudarshan, Database System Concepts, McGraw Hill.

References:

1. Martin Kleppmann, Designing Data-Intensive Applications, O'Reilly Media.
2. Kyle Banker, MongoDB in Action, Manning Publications.
3. Rajkumar Buyya, Mastering Cloud Computing, McGraw Hill.

Online References:

Sr. No.	Website Name
1.	https://www.coursera.org/specializations/big-data#courses
2.	https://www.digimat.in/nptel/courses/video/106106169/L01.html
3.	https://www.coursera.org/learn/nosql-databases#syllabus

Assessment:**Internal Assessment (IA) for 40 marks:**

- IA will consist of Two Compulsory Internal Assessment Tests of 20 marks each. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 60% to 50% of syllabus content must be covered in Second IA Test

End Semester Examination for 60 Marks**Question paper format**

- Question Paper will comprise of a total of six questions each carrying 15 marks Q.1 will be compulsory and should cover maximum contents of the syllabus
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of Four questions needs to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20361132	Industrial Robotics	3	-	-	3	-	-	3

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
20361132	Industrial Robotics	20	20	40	60	2	--	--	100

Rationale:

This course integrates industrial robotics systems used in automation and manufacturing. It provides understanding of robot operations, material handling and vision systems control techniques. The course enables students to analyze and design automated systems, supporting modern industrial requirements in robotics and automation.

Course Objectives:

1. To interpret the fundamentals of industrial robots and robotic material handling systems.
2. To analyze robotic vision systems and visual perception techniques used in automation.
3. To analyze industrial applications of robots in manufacturing processes.
4. To evaluate gripper mechanisms, robot selection criteria, and impact of robotization.
5. To Demonstrate basic robot programming concepts using robot programming commands, subroutines, and robotic programming languages such as VAL, AML, Python, and ROS.
6. Explain recent trends in robotics including AI applications, mobile robot locomotion, safety standards, socio-economic aspects, and emerging developments in robotic systems.

Course Outcomes:

1. Interpret robot configurations, workspace, and robotic material handling concepts.
2. Analyze robotic vision systems and object recognition techniques used in automation.
3. Analyze industrial robotic applications in manufacturing processes.
4. Evaluate gripper design, robot selection parameters, and economic aspects of robotization.
5. Analyze provide exposure to robot programming methods, robotic programming languages, and software tools used in industrial robotics and automation.
6. Analyze modern trends in robotics including artificial intelligence, mobile robotics, robot safety standards, and socio-economic aspects of automation.

DETAILED SYLLABUS:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Basic concepts of Engineering Mathematics and Engineering Mechanics for understanding motion, force, and load analysis; Manufacturing Processes for industrial applications of robots; and fundamentals of Electrical, Electronics, and Control Systems for understanding sensors, actuators, drives, and automation systems.	00	--
I	Fundamentals of Industrial Robots and Material Handling	Fundamentals of Industrial Robots: Classification and types of industrial robots based on configuration such as Cartesian, cylindrical, spherical, articulated, and SCARA robots, classification based on application and degree of freedom. Concept of robot payload and load handling capacity, robot work envelope and reach with basic representation and significance in industrial applications. Robotic Material Handling: General considerations in robotic material handling including part characteristics, handling requirements, and system flexibility. Machine loading and unloading using robots, CNC machine tool loading and unloading operations. Concept of robot-centered manufacturing cell including layout, components, and advantages in automated production systems. #Self-learning topics: Comparative study of manual and robotic material handling systems, industrial case studies of robot-centered manufacturing cells, demonstration videos of robotic loading and unloading systems.	06	CO1
II	Robotic Vision Systems and Visual Perception	Robotic Vision Systems: Introduction to robotic vision systems and their importance in automation and quality control. Image representation including pixels, resolution, and grayscale concepts. Basic image processing techniques such as filtering, thresholding, and segmentation. Visual Perception: Object recognition techniques used in industrial applications, depth measurement methods including stereo vision and structured light. Image data compression techniques and their relevance in real-time vision systems. Visual inspection systems used in manufacturing for defect detection and quality assurance. Software considerations including image acquisition, processing, and integration with robotic systems. #Self-learning topics: Industrial case studies of vision-guided robots, comparative overview of 2D and 3D vision systems, introduction to AI-based vision techniques.	06	CO2

III	Industrial Applications of Robots	Robotic Applications in Manufacturing: Overview of industrial robot applications in manufacturing industries. Robots in continuous arc welding including basic principle, system components, and advantages. Robots in spot welding including application requirements and industrial usage. Process Applications: Robots in spray painting including process characteristics, uniform coating requirements, and safety aspects. Robots in assembly operations including part handling, precision tasks, and automation challenges. Robots in cleaning, deburring, and surface finishing operations. Special Applications: Robots for underwater operations and hazardous environments including nuclear and chemical industries, highlighting safety and operational advantages. #Self-learning topics: Case studies of robotic welding and painting systems, comparison of manual and robotic processes, videos of robotic assembly lines in automotive industries.	06	CO3
IV	Grippers, Robot Selection, and Impact of Robotization	Grippers and End Effectors: Basic principles of gripping and gripper force analysis. Design considerations for grippers including shape, size, material, and application requirements. Types of grippers such as mechanical, vacuum, magnetic, and special-purpose grippers. Multi-degree-of-freedom grippers and their applications in complex handling tasks. Robot Selection and Performance: Selection criteria for industrial robots including payload, accuracy, repeatability, speed, and working environment. Performance evaluation parameters and testing methods for industrial robots. Economics and Impact: Economic aspects of robotization including cost-benefit analysis, productivity improvement, and return on investment. Impact of robotization on industry, workforce, and society. #Self-learning topics: Overview of soft and adaptive grippers, human-robot collaboration concepts, ethical and social implications of automation.	06	CO4
V	Programming and Languages for Robotics:	Robot Programming: Methods of robot programming, WAIT, SIGNAL and DELAY commands, subroutines, Programming Languages: Generations of Robotic Languages, Introduction to various types such as VAL, RAIL, AML, Python, ROS etc., Development of languages since WAVE till ROS.	07	CO5

VI	Trends in Robotics:	Socio-Economic aspect of robotisation. Economical aspects for robot design, Safety for robot and standards, Introduction to Artificial Intelligence, AI techniques, Need and application of AI, mobile robotics, New trends & recent updates in robotics. Mobile Robot locomotion: Types of locomotion, hopping robots, legged robots, wheeled robots, stability, manoeuvrability, controllability	07	CO6
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(Note: No questions should be asked on self-learning topics)

Text Books:

1. Industrial Robotics: Technology, Programming and Applications, by Mikell P. Groover, McGraw Hill Education.
2. Robotics: Control, Sensing, Vision and Intelligence, by K. S. Fu, R. C. Gonzalez and C. S. G. Lee, McGraw Hill.
3. Hydraulics and Pneumatics: A Technician's and Engineer's Guide, by Andrew Parr, Butterworth-Heinemann.
4. Introduction to Hydraulics and Pneumatics, by S. Ilango and V. Soundararajan, PHI Learning.
5. Fundamentals of Pneumatics and Hydraulics, by Md. Abdus Salam, Springer.
6. Industrial Hydraulics and Pneumatics, by P. B. Pawar, Sankalp Publication.

References:

1. Introduction to Robotics: Mechanics and Control, by John J. Craig, Pearson Education.
2. Industrial Robotics, by Ganesh S. Hegde, Laxmi Publications.
3. Oil Hydraulics and Pneumatics, by S. R. Majumdar, McGraw Hill Education.
4. Fluid Power with Applications, by Anthony Esposito, Pearson Education.
5. Fundamentals of Fluid Power Control, by John Watton, Cambridge University Press.
6. Principles of Hydraulic Systems Design, by Peter Chapple, Butterworth-Heinemann.
7. Pneumatic Control for Industrial Automation, by Peter Rohner.
8. Fluid Power Circuits and Controls, by John S. Cundiff, CRC Press.
9. Hydraulics and Electro-Pneumatics Training Manuals, by Festo Didactic, Festo Didactic GmbH.

Online References:

Sr. No.	Website Name
3.	NPTEL – Oil Hydraulics and Pneumatics: https://nptel.ac.in/courses/112104118
4.	NPTEL – Industrial Hydraulics and Automation: https://nptel.ac.in/courses/112104284
3.	NPTEL – Hydraulic Engineering: https://nptel.ac.in/courses/105105183
4.	NPTEL – Fundamentals of Industrial Oil Hydraulics and Pneumatics: https://nptel.ac.in/courses/112105046

5.	NPTEL – Introduction to Robotics: https://nptel.ac.in/courses/112101098
6.	NPTEL – Industrial Robotics: Theories for Implementation: https://nptel.ac.in/courses/112105319
7.	FluidSIM (Festo) – Pneumatics and Hydraulics Simulation Software: https://www.fluidsim.com

Assessment:

Internal Assessment (IA) for 40 marks:

- IA will consist of Two Compulsory Internal Assessment Tests of 20 marks each. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 60% to 50% of syllabus content must be covered in Second IA Test

End Semester Examination for 60 Marks

Question paper format

- Question Paper will comprise of a total of six questions each carrying 15 marks Q.1 will be compulsory and should cover maximum contents of the syllabus
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of Four questions needs to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20361133	Internet of Things (IOT)	3	-	-	3	-	-	3

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
20361133	Internet of Things (IOT)	20	20	40	60	2	--	--	100

Rationale:

This course aims to equip students with the theoretical foundation and practical knowledge to design, implement, and manage interconnected smart systems. Understanding how physical devices, sensors, and actuators are integrated via the internet is essential for developing intelligent and autonomous robotic solutions and automated industrial processes, thus aligning with the core requirements of the branch.

Course Objectives:

Upon successful completion of the course, students will be able to:

1. To understand IoT Characteristics and Conceptual
2. To comprehend network architecture and design of IoT.
3. To understand smart objects in IoT
4. To correlate the connection of smart objects and IoT access technologies
5. To explore network layer and application layer protocols for IoT.
6. To explore IoT security aspect

Course Outcomes:

1. Analyze the IoT Characteristics and Conceptual Framework
2. Analyze between the levels of the IoT architectures
3. Interpret sensor network and its components
4. Analyze the IoT access technologies
5. Analyze and evaluate various protocols at network layer and application layer for IoT
6. Analyze and evaluate security issues in IoT and risk analysis structure

DETAILED SYLLABUS: total six module for each subject (13 Weeks)

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Python programming, C programming language, Computer Networks, Ports, Timers, Programming of controller, How to use IDE to write code of microcontroller, TCP-IP protocol stack, Setup the IDE, Arduino Libraries for sensors.	0	
I	Introduction to IoT	Introduction and application to Internet of Things: need of IoT , defining IoT, Characteristics of IoT, Conceptual Framework of IoT, Communication models & APIs, Basics of networking Communication protocol, wireless sensor networks, IoT Challenges, IoT protocol vs Web Protocol stack Hardware and software development tools for –NodeMCU, ESP32, Raspberry Pi	6	CO 1, CO6
II	Design and Architecture of IoT Network	2.1 IoT World Forum (IoTWF) Standardized Architecture: Layer 1-7, A Simplified IoT Architecture, The Core IoT Functional Stack::Layer 1-3, Analytics Versus Control Applications, Data Versus Network Analytics Data Analytics Versus Business Benefits, Smart Services, 2.2 Types of edges and its working (i.e Fog Computing , Edge Computing ,The Hierarchy of Edge, Fog, and Cloud)	6	CO 2
III	Smart Objects IoT	3.1 Micro-Electro-Mechanical Systems (MEMS) Smart Objects: A Definition, Trends in Smart Objects, 3.2 Sensor Networks, Wireless Sensor Networks (WSNs), Communication Protocols - WSN, RFID, NFC, IEEE 802.15.4, Zigbee and IPv6 Technologies for IoT.	7	CO 3
IV	Connecting Smart Objects	4.1 Selection of Range, Frequency Bands, Power Consumption, Topology, Constrained Devices , Constrained-Node Networks, Data Rate and Throughput , Latency and Determinism 4.2 Technologies in IoT: Standardization and Alliances, Physical Layer, MAC Layer, Topology, Security and Conclusion of IEEE 802.15.4 , IEEE 802.15.4g and 802.15.4e ,IEEE 1901.2a ,IEEE 802.11ah , 4.3 LoRa WAN, and NB-IoT and Other LTE Variations , LTE Cat 0 , LTE- M, NB-IoT	7	CO 4
V	IoT Application protocols and Cloud	5.1 IoT Network Layer and Application Layer Protocols, CoAP, MQTT, AMQP 5.2 Cloud Computing Concept, Grid/SOA and Cloud Computing, Cloud Middleware, NIST's SPI Architecture and Cloud Standards, The Cloud of Things--The Internet of Things and Cloud Computing,	6	CO 5

VI	Securing IoT	6.1 Common Challenges in OT Security : Erosion of Network Architecture, Pervasive, Legacy Systems, Insecure Operational Protocols like Modbus, DNP3 ,ICCP ,OPC , (IEC) Protocols, Device Insecurity 6.2 Security Knowledge: IT and OT Security Practices and Systems Vary, The Purdue Model for Control Hierarchy, OT Network Characteristics Impacting Security, Security Priorities: CIA, Security Focus 6.3 Formal Risk Analysis Structures: OCTAVE and FAIR, FAIR The Phased Application of Security in an Operational Environment, Secured Network Infrastructure and Assets, Deploying Dedicated Security Appliances, Higher-Order Policy Convergence and Network Monitoring	7	CO 6
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Text Books:

1. "The Internet of Things: Enabling Technologies, Platforms, and Use Cases" by Pethuru Raj and Anupama C. Raman.
2. "IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things" by David Hanes, Gonzalo Salgueiro, Robert Barton, Jerome Henry, and Patrick Grossetete.
3. "Designing the Internet of Things" by Adrian McEwen and Hakim Cassimally.
4. Building an IoT: A Step-by-Step Guide to Designing, Prototyping, and Implementing an IoT Solution" by Michael Massaros.
5. "Internet of Things (IoT): Principles, Paradigms and Applications" by S. Misra, A. Mukherjee, and A. Roy.
6. David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry, IoT Fundamentals Networking Technologies, Protocols, and Use Cases for the Internet of Things CISCO.
7. Arsheep Bahga (Author), Vijay Madisetti, Internet Of Things: A Hands-On Approach Paperback, Universities Press, Reprint 2020

References:

1. Pethuru Raj, Anupama C. Raman, The Internet of Things: Enabling Technologies, Platforms, and Use Cases by , CRC press,
2. Raj Kamal, Internet of Things, Architecture and Design Principles, McGraw Hill Education, Reprint 2018.
3. Perry Lea, Internet of Things for Architects: Architecting IoT solutions by implementing sensors, communication infrastructure, edge computing, analytics, and security, Packt Publications, Reprint 2018.
4. Amita Kapoor, "Hands on Artificial intelligence for IoT", 1st Edition, Packt Publishing, 2019.

Recommended SWAYAM/NPTEL Courses

Sr. No.	Recommended SWAYAM/NPTEL Courses
1.	Introduction to Internet of Things (IIT Kharagpur/IIT Madras) nptel.ac.in/courses/106105166
2.	Design for Internet of Things (Prof. Prabhakar, IISc Bangalore) Design for internet of things - Course
3.	Internet of Things: Design Concepts and Use Cases (NITTTR, Multiple Instructors) Internet of Things: Design Concepts and Use Cases - Course
4.	Embedded Systems Design (Prof. Anupam Basu, IIT Kharagpur) nptel.ac.in/courses/106105159

Assessment:

Internal Assessment (IA) for 40 marks:

- IA will consist of Two Compulsory Internal Assessment Tests of 20 marks each. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 60% to 50% of syllabus content must be covered in Second IA Test

End Semester Examination for 60 Marks

Question paper format

- Question Paper will comprise of a total of six questions each carrying 15 marks Q.1 will be compulsory and should cover maximum contents of the syllabus
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of Four questions needs to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20361141	Biomedical Instrumentation	3	-	-	3	-	-	3

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
20361141	Biomedical Instrumentation	20	20	40	60	2	--	1	100

Course Objectives:

1. To make students know the physiology of cardiovascular system, respiratory system, nervous and muscular systems, find problems related to them and specify the corrective action to be taken.
2. To make students understand and characterize the origin of bio-potentials and inspect common biomedical signals by their characteristics features and design bio-potential amplifiers.
3. To operate ECG, EEG, EMG and BP, to monitor measure and analyze heart activities.
4. To use different imaging Techniques
5. To know various Life supporting Instruments
6. To apply safety practices while using biomedical instruments

Course Outcomes:

1. Apply various Bio-potential and their specifications and can perform measurements.
2. Analyze various Physiological systems and to identify their parameters and related measurements.
3. Analyze principle and working of various cardiovascular parameters, their measurements, applications.
4. Analyze various medical imaging techniques based on the principles and concepts involved in them.
5. Analyze different life support instruments and their applications.
6. Analyze the significance of electrical safety in biomedical measurement.

DETAILED SYLLABUS: total six module for each subject (13 Weeks)

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
I	Physiological System:	Cardiovascular system- Structure of Heart, electrical and Mechanical activity of heart, ECG measurement, cardiac arrhythmias, heart sounds, ECG amplifier design Respiratory system- physiology, measurement of respiratory related parameters, Spirometer. Nervous system-Nerve cell neural communication, Nerve-muscle physiology, Generation of EEG, its characteristics, normal & abnormal EEG. EEG amplifier. Muscular system: Generation of EMG signal and its measurement, EMG amplifier.	10	CO1
II	Bio-Potential Measurement:	Structure of Cell, Origin of Bio-potential, electrical activity of cell and its characteristics and specifications. Measurement of RMP, AP, Electrode-Electrolyte interface, Biopotential electrodes: types	04	CO2
III	Cardiovascular measurement:	Phonocardiography, Blood Pressure Measurement: Direct and indirect, Blood Flow meters: Electromagnetic and Ultrasonic, Thermal Convection Methods, Cardiac Output Measurement- Fick method, Dye-dilution and Thermo-dilution method, Blood Volume measurement-Plethysmography	08	CO3
IV	Imaging Techniques:	X-ray tube- construction, working, applications. CT scan- CT number, block diagram, scanning system, applications, MRI- working, applications. Ultrasound Imaging- working, modes of scanning, application Magnetic Resonance Imaging (MRI) – Principle, block diagram and application.	08	CO4
V	Life support Instruments	Pacemaker- Types, modes of packing, applications, Defibrillators-Ac and DC with applications, Heart-Lung machine its importance during surgery, Hemodialysis system, precautions to be taken during dialysis. Ventilator system: its important parameters for monitoring	09	CO5
VI	Electrical Safety	Physiological effects of electrical current, Shock Hazards from electrical equipment and methods of accident prevention.	02	CO6
		Total Hr.	39	

* A Hospital Visit is recommended.

Text Books:

- 1 Leslie Cromwell, —Biomedical Instrumentation and Measurements, 2nd Edition, Pearson Education, 1980.
2. R. S. Khandpur, —Biomedical Instrumentation, TMH, 2004
3. John G. Webster, —Medical Instrumentation, John Wiley and Sons, 4th edition, 2010.
4. Human Physiology- The Mechanism of Body Function By Vander, Sherman, TMH Ed. 1981.

References:

1. Biomedical Instrumentation, Dr. M. Arumugam , Anuradha Publications,
- 2 Richard Aston, “Principles of Biomedical Instrumentation and Instruments”, PH, 1991.
3. John E Hall, Gyton’s Medical Physiology, 12th edition, 2011
4. Joseph J. Carr and John M. Brown, “Introduction to Biomedical Equipment Technology”, PHI/Pearson Education, 4th edition, 2001.

Assessment:

Internal Assessment (IA) for 40 marks:

- IA will consist of Two Compulsory Internal Assessment Tests of 20 marks each. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 40% to 50% of syllabus content must be covered in Second IA Test

End Semester Examination for 60 Marks

Question paper format

- Question Paper will comprise of a total of six questions each carrying 15 marks Q.1 will be compulsory and should cover maximum contents of the syllabus
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of Four questions needs to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20361142	Micro Electro Mechanical System	3	-	-	3	-	-	3

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
20361142	Micro Electro Mechanical System	20	20	40	60	2	--	--	100

Rationale:

Micro Electro Mechanical Systems (MEMS) integrate mechanical components, sensors, actuators, and electronics at the micro-scale using advanced microfabrication techniques. MEMS technology is widely used in automotive systems, consumer electronics, biomedical devices, and industrial automation. This course introduces students to the fundamentals of MEMS materials, fabrication processes, sensors, actuators, and applications. It helps students understand the design and working principles of microscale devices, enabling them to explore modern smart systems. The course promotes interdisciplinary learning and awareness of emerging micro-scale technologies used in modern engineering applications.

Course Objectives:

1. Explain the principles, characteristics, and scaling effects associated with Micro Electro Mechanical Systems (MEMS).
2. Describe MEMS materials and the microfabrication processes such as lithography, deposition, etching, and micromachining used in MEMS manufacturing.
3. Analyze the operating principles of MEMS-based microsensors and micro actuators used in automation and robotic systems.
4. Evaluate the role of MEMS technology in modern engineering applications including robotics, biomedical devices, and smart systems.
5. Describe different types of micro actuators such as electrostatic, piezoelectric, thermal, electromagnetic, and SMA-based actuators used in precision systems.
6. Analyze MEMS applications and case studies related to robotics, automation, medical systems, and microfluidic devices, along with the impact of materials and fabrication processes on MEMS performance.

Course Outcomes:

1. Analyze the fundamental concepts, characteristics, and scaling effects of MEMS devices.
2. Analyze and evaluate MEMS materials and microfabrication processes used in the development of microsystems.
3. Apply appropriate microfabrication techniques for the development of MEMS devices.
4. Analyze the working principles of MEMS-based microsensors and micro actuators used in automation and robotic applications.

5. Evaluate MEMS technologies considering material selection, fabrication methods, and performance requirements for engineering applications.
6. Design MEMS-based solutions for selected applications in robotics, automation, or biomedical systems.

DETAILED SYLLABUS:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Introduction to micro-scale engineering. Basic concepts of semiconductor materials. Overview of sensors and actuators. Fundamentals of thin film technology. Importance and scope of MEMS in modern technology (Note: There will be no questions from the prerequisite in the theory examination)	00	
I	Introduction to Micro Electro Mechanical System	Overview of Micro Electro Mechanical Systems (MEMS) and Microsystems Technology. Unique characteristics of MEMS: miniaturization, multifunctionality, batch fabrication, low power consumption, high sensitivity. Typical MEMS and microsystem products used in modern engineering systems. Scaling effects in MEMS: surface-to-volume ratio, mechanical and electrical implications of miniaturization. Scaling laws in microsystems design, Applications of MEMS in Automation and Robotics: inertial sensors for navigation, MEMS sensors in drones, robotic motion sensing, industrial automation systems #Self-learning topics: Historical evolution of MEMS technology. Introduction to Nano Electro Mechanical Systems (NEMS), MEMS devices used in smartphones, wearable devices, and IoT systems, MEMS-based sensors used in autonomous vehicles and drone stabilization.	06	CO1
II	Materials for MEMS and Manufacturing	Material requirements for MEMS devices, Structure and properties of silicon and other MEMS materials, Silicon wafer processing and wafer preparation, Bulk micromachining and Surface micromachining, Wafer bonding techniques, Thin film deposition methods: PVD and CVD, Lithography techniques in MEMS fabrication, Etching processes: Wet etching and Dry etching #Self-learning topics: Emerging MEMS materials: polymers, ceramics and graphene, SOI wafers (Silicon-on-Insulator) technology, Cleanroom facilities used in MEMS fabrication	08	CO2

III	Microfabrication Methods	LIGA process and its applications, Moulding techniques in MEMS fabrication, Soft lithography techniques, Polymer processing for MEMS devices, Thick-film processing techniques, Low Temperature Co-fired Ceramic (LTCC) processing, Comparison of microfabrication techniques used in MEMS device manufacturing #Self-learning topics: 3D printing in microfabrication, Nanoimprint lithography, Microfabrication techniques used in micro-robotics.	07	CO3
IV	MEMS Components – Micro Sensors	Classification and working principles of MEMS microsensors, Acoustic wave microsensors, Optical microsensors, Pressure microsensors used in industrial control and robotics, Thermal microsensors, Acceleration microsensors (MEMS accelerometers) used in robotic motion sensing and stabilization, Introduction to chemical microsensors (Focus should be on MEMS design and working principles rather than basic sensor theory) #Self-learning topics: Biomedical microsensors and biosensors, MEMS sensors used in robotic perception systems, MEMS based gas sensors for environmental monitoring	08	CO4
V	Micro-Actuators	Fundamentals of micro actuators, Electrostatic micro actuators, Piezoelectric micro actuators, Thermal micro actuators, Electromagnetic micro actuators, Shape Memory Alloy (SMA) micro actuators, MEMS based micro valves and micro pumps used in precision control systems #Self-learning topics: Micro actuators used in micro-robotics and medical devices, MEMS actuators used in optical switching systems, Artificial muscle actuators	06	CO5
VI	MEMS Applications and Case Studies	Case studies of MEMS applications in engineering systems, Impact of materials, fabrication processes and design on MEMS performance, MEMS devices used in robotics and automation systems, Micropumps and microfluidic devices in engineering and medical applications #Self-learning topics: MEMS devices used in space applications, MEMS based lab-on-chip systems, Recent research trends in MEMS for autonomous systems	04	CO6

(Note: No questions should be asked on self-learning topics)

Text Books:

1. MEMS and Microsystems: Design and Manufacture, by Tai-Ran Hsu, Tata McGraw-Hill Publishing Company Ltd.
2. Foundations of MEMS, by Chang Liu, Pearson Education.
3. An Introduction to Microelectromechanical Systems Engineering, by Nadim Maluf and Kirt Williams, Artech House.

References:

1. Fundamentals of Microfabrication and Nanotechnology, by Marc J. Madou, CRC Press, 2011, ISBN: 9780849331800.
2. Micromachined Transducers Sourcebook, by Gregory Kovacs, WCB McGraw-Hill, Boston, 1998, ISBN: 9780071164627.
3. Micromechanical Transducers: Pressure Sensors, Accelerometers, and Gyroscopes, by M.H. Bao, Elsevier, New York, 2000, ISBN: 9780444505583.
4. Microsystem Design, by Stephen D. Senturia, Springer Publication, 2000, ISBN: 9780792372462.
5. Microsensors: Principles and Applications, by Julian W. Gardner, John Wiley & Sons, 1994, ISBN: 9780471941361.
6. MEMS: Introduction and Fundamentals, by Mohamed Gad-el-Hak, CRC Press.
7. MEMS and NEMS: Systems, Devices and Structures, by Sergey Edward Lyshevski, CRC Press.

Online References:

Sr. No.	Website Name
1	NPTEL – MEMS and Microsystems: https://nptel.ac.in/courses/117105082
2	NPTEL – Advanced Manufacturing Process for Micro System Fabrication: https://nptel.ac.in/courses/112104162
3	NPTEL – BioMEMS and Microfluidics: https://nptel.ac.in/courses/112104181
4	NPTEL – Microsensors and Nanosensors: https://onlinecourses.nptel.ac.in/noc25_ee177/preview
5	NPTEL – Introduction to Micro Sensors: https://nptel.ac.in/courses/108106165

Assessment:**Internal Assessment (IA) for 40 marks:**

- IA will consist of Two Compulsory Internal Assessment Tests of 20 marks each. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 60% to 50% of syllabus content must be covered in Second IA Test

End Semester Examination for 60 Marks**Question paper format**

- Question Paper will comprise of a total of six questions each carrying 15 marks Q.1 will be compulsory and should cover maximum contents of the syllabus
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of Four questions needs to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20361143	Hydraulics and Pneumatics	3	-	-	3	-	-	3

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
20361143	Hydraulics and Pneumatics	20	20	40	60	2	--	--	100

Rationale:

This course introduces hydraulic systems used in automation and robotics, covering fluid power principles, components, and circuit design. It also includes machinery diagnostics for condition monitoring and fault detection. The course enables students to develop skills in analysis, operation, and maintenance of hydraulic systems, aligning with modern industrial requirements.

Course Objectives:

1. To interpret principles of fluid power, fluid properties, and governing laws for hydraulic and pneumatic systems.
2. To analyze the performance characteristics of hydraulic pumps and actuators for appropriate application selection.
3. To develop hydraulic circuits using control valves and standard symbols for industrial operations
4. To analyze electro-hydraulic systems and control logic for automation applications.
5. To analyze pneumatic systems, components, and control valves used in automation.
6. To develop pneumatic and electro-pneumatic circuits for control and sequencing applications.

Course Outcomes:

1. Interpret fluid power principles, fluid behaviour, and system performance using fundamental laws and equations.
2. Analyze hydraulic pumps and actuators based on performance parameters and application requirements.
3. Develop hydraulic circuits incorporating valves and control strategies for industrial applications.
4. Analyze electro-hydraulic systems and associated control mechanisms for automation tasks.
5. Evaluate hydraulic system performance, maintenance strategies, and safety practices in industrial environments.
6. Develop pneumatic and electro-pneumatic circuits for industrial control and sequencing.

DETAILED SYLLABUS:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Fundamentals of Engineering Mechanics (force, work, energy), Basic Physics (pressure, fluid properties) Introductory Electrical Engineering (for electro-hydraulic systems), Basic understanding of Automation/Control Systems (desirable)	00	--
I	Fundamentals of Fluid Power Systems	Hydraulic and pneumatic systems: working principles Comparison of hydraulic vs pneumatic systems Advantages and limitations of fluid power systems in automation Properties of Fluids: Physical properties: density, viscosity, Bulk modulus and compressibility, Vapor pressure and cavitation concept, Types of hydraulic fluids: HFA, HFB, HFC, HFD and their applications Fluid Statics and Basic Laws: Hydrostatic pressure and its measurement, Pascal's Law and its application in hydraulic systems Fluid Dynamics and Energy Relations: Continuity equation, Bernoulli's equation, General energy equation (including pump head and losses) Flow Regimes and Losses: Laminar and turbulent flow, Reynolds number and its significance, Major and minor losses in pipes, Darcy-Weisbach equation for head loss (application level, no derivation) System Performance and Pneumatic Fundamentals: Hydraulic power calculation, System efficiency, Introduction to pneumatic systems, Properties of air and compressibility, Standard air flow rates #Self-learning topics: Industrial applications of hydraulic and pneumatic systems in robotics, Cavitation effects and prevention techniques, Case study: Fluid power system in automated manufacturing, Comparison of energy efficiency in hydraulic vs pneumatic systems	08	CO1
II	Hydraulic Pumps and Actuators	Hydraulic Pumps: Classification of pumps: positive displacement and non-positive displacement, Gear pump, vane pump, piston pump (construction and working), Pump performance parameters: discharge, slip, efficiency Pump Characteristics and Selection: Volumetric and overall efficiency, Pump performance curves (basic understanding), Selection of pumps based on system requirements Hydraulic Actuators: Linear actuators: single acting and double acting cylinders, Rotary actuators: hydraulic motors, Force, speed, and power relations	07	CO2

		Actuator Performance: Load analysis, Cylinder cushioning, Applications in automation systems #Self-learning topics: Comparison of pump types for industrial applications, Failure modes of hydraulic pumps and actuators, Case study: Actuator selection in robotic systems		
III	Hydraulic Control Valves and Circuits	Hydraulic Control Valves: Direction control valves (2/2, 3/2, 4/2, 4/3), Pressure control valves: relief, reducing, sequence valves, Flow control valves: throttle, pressure-compensated Hydraulic Circuit Symbols: ISO symbols and representation, Reading and interpretation of hydraulic circuits Basic Hydraulic Circuits: Single acting and double acting cylinder circuits, Regenerative circuits, Pressure sequencing circuits Speed Control Circuits: Meter-in and meter-out circuits, Load effects on speed control #Self-learning topics: Design of hydraulic circuits for industrial machines, Circuit simulation using Fluid SIM or similar tools, Case study: Hydraulic press circuit	07	CO3
IV	Electro-Hydraulic Systems	Electro-Hydraulic Control Systems: Need and advantages, Integration of electrical and hydraulic systems Electrical Components in Hydraulic Systems: Solenoid-operated valves, Relays and switches, Limit switches and sensors (introductory) Control Logic: Relay-based control circuits, Sequence control of actuators Advanced Electro-Hydraulic Systems: Proportional and servo valves, Closed-loop control systems, Feedback mechanisms and controllers, PLC-based electro-hydraulic control #Self-learning topics: PLC ladder diagrams for hydraulic applications, Case study: Automated hydraulic system, Study of electro-hydraulic systems in robotics	04	CO4
V	Pneumatic Systems and Components	Pneumatic System Fundamentals: Properties of air including compressibility, density, and moisture content. Working principle of pneumatic systems and their advantages and limitations in automation. Air Supply System: Types of compressors such as reciprocating and rotary compressors, their working principles and specifications. Air preparation units including filter, regulator, and lubricator (FRL unit) and their role in maintaining system performance. Complete pneumatic system layout including air supply, control elements, and actuators. Actuators and Valves: Classification and working of pneumatic cylinders including single-acting and double-acting cylinders. Directional	06	CO5

		control valves including number of ways and positions, valve characteristics, and actuation methods. Pressure control valves, quick exhaust valves, sequence valves, and flow control valves. Logic Valves: Pneumatic logic elements including AND, OR, and time delay valves and their role in automation control. #Self-learning topics: Industrial applications of pneumatic systems, selection of compressors and actuators for automation tasks		
VI	Pneumatic Circuit Design and Electro-Pneumatics	Pneumatic Circuit Design: Speed control circuits using meter-in and meter-out methods. Design of pneumatic logic circuits using AND and OR functions for automation. Sequential circuit design methods including displacement-step diagram, cascade method, shift register method, and stepper sequencer method. Sensing and Control: Position sensing in pneumatic cylinders using limit switches and sensors. Pressure sensing and use of roller lever valves. Signal overlapping and its analysis in sequential circuits. Electro-Pneumatics: Electrical control components including push buttons, limit switches, pressure switches, solenoid valves, relays, timers, and P/E converters. Memory circuits including dominant ON/OFF control, latching circuits, and counting circuits. Circuit Optimization and Energy Efficiency: Methods to optimize pneumatic circuits for reduced air consumption and improved system efficiency. Use of pressure regulators, flow control valves, and air-saving techniques. Minimization of leakage and pressure losses. Performance analysis of circuits considering speed, load, and energy consumption. Troubleshooting: Common faults in pneumatic systems and systematic troubleshooting techniques. #Self-learning topics: PLC-based pneumatic automation examples, industrial case studies of electro-pneumatic systems.	07	CO6

(Note: No questions should be asked on self-learning topics)

Text Books:

1. Fluid Power with Applications, by Anthony Esposito, Pearson Education.
2. Hydraulics and Pneumatics: A Technician's and Engineer's Guide, by Andrew Parr, Butterworth-Heinemann.
3. Introduction to Hydraulics and Pneumatics, by S. Ilango and V. Soundararajan, PHI Learning.
4. Hydraulics and Pneumatics, by Jagadeesha T., I.K. International Publishing House.

5. Fundamentals of Pneumatics and Hydraulics, by Md. Abdus Salam, Springer.
6. Industrial Fluid Power, by John Watton, Cambridge University Press.

References:

1. Oil Hydraulics Systems: Principles and Maintenance, by S. R. Majumdar, McGraw Hill Education.
2. Pneumatic Systems: Principles and Maintenance, by S. R. Majumdar, McGraw Hill Education.
3. Fundamentals of Pneumatics and Hydraulics, by Md. Abdus Salam, Springer.
4. Hydraulics and Pneumatics, by Jagadeesha T., I.K. International Publishing House.
5. An Introduction to Predictive Maintenance, by R. Keith Mobley, Elsevier.
6. Hydraulics and Electro-Hydraulics Training Manuals, by Festo Didactic, Festo Didactic GmbH.

Online References:

Sr. No.	Website Name
1.	NPTEL – Oil Hydraulics and Pneumatics: https://nptel.ac.in/courses/112104118
2.	NPTEL – Industrial Hydraulics and Automation: https://nptel.ac.in/courses/112104284
3.	NPTEL – Hydraulic Engineering: https://nptel.ac.in/courses/105105183
4.	NPTEL – Fundamentals of Industrial Oil Hydraulics and Pneumatics: https://nptel.ac.in/courses/112105046
5.	FluidSIM (Festo) – Pneumatics and Hydraulics Simulation Software: https://www.fluidsim.com

Assessment:

Internal Assessment (IA) for 40 marks:

- IA will consist of Two Compulsory Internal Assessment Tests of 20 marks each. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 60% to 50% of syllabus content must be covered in Second IA Test

End Semester Examination for 60 Marks

Question paper format

- Question Paper will comprise of a total of six questions each carrying 15 marks Q.1 will be compulsory and should cover maximum contents of the syllabus
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of Four questions needs to be answered

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2036115	Robotic Control System Lab	--	02	-	--	01	-	01

Course Code	Course Name	Examination Scheme						
		Theory Marks				Term Work	Practical/ Oral	Total
		Internal assessment			End Sem. Exam			
		Test1	Test 2	Avg. of 2 Tests				
2036115	Robotic Control System Lab	--	--	--	--	25	25	50

Course Objectives:

1. To expose students to the practical aspects of dynamic modeling and simulation of robotic manipulators.
2. To provide opportunities for implementing nonlinear control techniques for robotic manipulators using suitable tools.
3. To develop experimental understanding of force control and interaction dynamics in manipulators.
4. To familiarize students with the kinematic modeling and control of steered robots through hands-on exercises.
5. To enable students to experiment with vision-based sensing and control approaches in robotics.
6. To strengthen practical knowledge of PID control strategies for single-link and planar 2R robotic manipulators.

Course Outcomes:

1. Apply the basic concepts of dynamic modeling to generate and analyze robot joint trajectories and dynamics of robotic manipulators.
2. Apply and analyze nonlinear control techniques to design stiffness and trajectory control for robotic manipulators.
3. Analyze and design kinematic and dynamic control strategies for multi-link manipulators using forward and inverse kinematics principles.
4. Apply and analyze kinematic models to design and implement control algorithms for mobile robots such as line follower and navigation systems.
5. Apply and estimate the performance of intelligent and vision-based control algorithms for mobile robotic systems.
6. Apply and analyze PID-based and other advanced control techniques for manipulators, and estimate performance through simulation or industrial case studies.

List of Experiments.

Sr. No	List of Experiments	LO Mapping	Hrs
01	Creating robot joint trajectories	LO1	02
02	Study Inverse and Forward dynamics of robots	LO1	02
03	Design stiffness control for 3-link planner manipulator	LO2	02
04	Design trajectory following controller for a given system.	LO3	02
05	Design trajectory following controller for a linear system or using linearized model of the system.	LO3	02
06	Study of Forward Kinematics of a 5R Robot Manipulator	LO3	02
07	Study of Inverse Kinematics of a 5R Robot Manipulator	LO3	02
08	Design of controller for Multilink Manipulator	LO3	02
09	Design of Line follower robot control.	LO4, LO5	02
10	Design of Navigation control of mobile robot using Neural Network algorithm.	LO4, LO5	02
11	Design Control a virtual robot using a joystick	LO6	02
12	Study workspace Analysis of a 6-axis robot	LO6	02
13	Design PID Controller for single link manipulator	LO6	02
14	Case study on any Industrial Robot	LO6	01

Sr No	List of Assignments	Hrs.
01	Mathematical representation of robot joint trajectories and parameters influencing motion profiles such as position, velocity, and acceleration in manipulators	02
02	Dynamic behavior of robotic manipulators including forward and inverse dynamics and their significance in motion control	02
03	Concept of stiffness control in multi-link manipulators and its role in interaction with environment during task execution	02
04	Trajectory tracking in robotic systems including characteristics of linear and nonlinear control approaches for motion control	02
05	Kinematic modeling of robotic manipulators including forward and inverse kinematics and their role in control of multi-link systems	02
06	Control strategies for mobile robots including line following and navigation using intelligent algorithms such as neural networks	02

Assessment:

Term Work: Term Work shall consist of at least 10 Experiments based on the above list. Also, Term work Journal must include at least 2 assignments.

Distribution of Marks for Term Work (25 Marks):

Laboratory Work	15 Marks
Assignments	05 Marks
Attendance	05 Marks

End Semester Practical/Oral Examination:

1. Pair of Internal and External Examiner should conduct practical / viva based on contents
2. Distribution of marks for practical/viva examination shall be as follows:
 - a. Practical performance 15 Marks
 - b. Viva 10 Marks
3. Evaluation of practical examination to be done based on the experiment performed and the output of the experiment during practical examination
4. Students work along with evaluation report to be preserved till the next examination

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2036116	Process Instrumentation and Control Lab	2	-	-	2	-	-	2

Course Code	Course Name	Examination Scheme						
		Theory Marks				Term Work	Practical/ Oral	Total
		Internal assessment			End Sem. Exam			
		Test 1	Test 2	Avg. of 2 Tests				
2036116	Process Instrumentation and Control Lab	--	--	--	--	25	25	50

Lab Objectives:

1. To make students to identify the elements of feedback control system for the given process.
2. To make students to identify the elements of feed-forward control system for the given process.
3. To make students to identify the elements of Cascade control system for the given process.
4. To select relevant Instrumentation for various unit operations in the process plant.
5. To measure the various process parameters for the given dryer system.
6. To analyze working of AC and DC servomotor, Stepper motor

Lab Outcomes:

1. Relate the working of different types of control actions.
2. Relate the working of different types of controllers and their tuning methods.
3. Analyze various control schemes and their application.
4. Apply the knowledge of different motors with applications.
5. Apply and analyze various process parameters for given boiler control.
6. Plot the thermal characteristics of the given type of heat exchanger.

List of Experiments.

Sr No	List of Experiments	LO	Hrs
01	Use ON-OFF controller for temperature control system.	LO 1	02
02	Response of proportional control for any unit operation	LO 1	02
03	Response of proportional + integral control for any unit operation	LO 1	02

04	Response of PID control for unit operation	LO 2	02
05	Feedback control loop, Feed forward control loop, Cascade control loop	LO 3	02
06	Operation of pneumatic/ hydraulic single acting cylinder with the help of directional control valve	LO 4	02
07	Operation of pneumatic/ hydraulic double acting cylinder with the help of directional control valve	LO 4	02
08	Operation and control of heat exchanger	LO 4	02
09	Boiler Instrumentation (Measure various process parameters using boiler)	LO 4	02
10	Demonstrate the operation & control scheme of dryer	LO 5	02
11	Angular position control using DC Servo system	LO 6	02
12	Use Stepper motor servo system component and measure its speed	LO 6	02
13	Use different servo component for controlling angular position of given DC servo motor	LO 6	02
14	Use different servo component for controlling angular position of given AC servo motor	LO 3	02
15	Identification of Various parts of control valve	LO 2	02

Sr No	List of Assignments	Hrs.
01	Dynamic behavior of first and second order systems. Interacting and non-interacting systems, Development of Mathematical Model for first & second order system	02
02	Compare PI,PD,PID controller, write output equations for same	02
03	Block diagrams of adaptive control and inferential control. Explain their working. List their applications.	02
04	Compare DC and AC servo motors.	02
05	State types of control valve. Draw their characteristics.	02
06	Explain working of tube heat exchanger. Working of Fluidised bed and Double drum dryer	02

Assessment:

Term Work: Term Work shall consist of at least 10 Experiments based on the above list. Also, Term work Journal must include at least 2 assignments.

Distribution of Marks for Term Work (25 Marks):

Laboratory Work	15 Marks
Assignments	05 Marks
Attendance	05 Marks

End Semester Practical/Oral Examination:

1. Pair of Internal and External Examiner should conduct practical / viva based on contents
2. Distribution of marks for practical/viva examination shall be as follows:
 - a. Practical performance 15 Marks
 - b. Viva 10 Marks
3. Evaluation of practical examination to be done based on the experiment performed and the output of the experiment during practical examination
4. Students work along with evaluation report to be preserved till the next examination

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20361171	Database Management System (DBMS) Lab		2	-	3	1	-	4

Course Code	Course Name	Examination Scheme						
		Theory Marks				Term Work	Practical/ Oral	Total
		Internal assessment			End m. Exam			
		Test1	Test 2	Avg. of 2 Tests				
20361171	Database Management System (DBMS) Lab	--	--	--	--	25	25	50

Lab Objectives:

1. To implement database concepts practically using SQL and NoSQL systems.
2. To perform data modeling, normalization, and schema design for real-life problems.
3. To write complex SQL queries and explore advanced database functionalities like triggers and stored procedures.
4. To understand transaction control and concurrency through simulation experiments.
5. To work with modern databases such as MongoDB and integrate SQL-NoSQL systems in mini-projects.
6. To practice secure database access and optimize performance using indexing and query analysis tools.

Lab Outcomes:

1. Design ER/EER models and implement normalized relational schemas.
2. Write optimized SQL queries, subqueries, joins, and triggers for data manipulation.
3. Implement and compare relational and NoSQL databases for specific use cases.
4. Apply transaction management, recovery, and concurrency control techniques.
5. Develop a mini-project integrating multiple databases and front-end interfaces.
6. Ensure database security and performance using indexing and optimization tools.

List of Experiments

Sr No	List of Experiments	LO	Hrs
01	Design an ER diagram for a real-world scenario (Library/College/Banking system)	LO1	2
02	Map the ER model to a relational schema and implement tables using SQL	LO1	2
03	Write a DDL and DML commands in SQL (Create table, drop, insert.....)	LO2	2

04	Write SQL queries using selection, projection, joins, and aggregate functions.	LO2	2
05	Implement nested and complex SQL queries including views and triggers.	LO2	2
06	Implement transaction control using COMMIT, ROLLBACK, and SAVEPOINT.	LO4	2
07	Simulate concurrency and recovery using log-based transaction management.	LO4	2
08	Explore indexing and query optimization using EXPLAIN or ANALYZE statements.	LO3	2
09	Install and work with a NoSQL database (MongoDB/Cassandra). Perform CRUD operations.	LO3	2
10	Implement database security mechanisms – user roles, privileges, and SQL injection prevention.	LO4	2
11	Implement Storage Service on Cloud using OpenStack	LO6	2
12	Mini Project: Develop a small end-to-end data-driven application (e.g., Student Management, E-commerce, or IoT Sensor Data).	LO5	2

Sr No	List of Assignments / Tutorials	Hrs
01	Design an ER/EER diagram and relational schema for a real-world system such as Library or College Database	2
02	Apply relational algebra operations and map them to equivalent SQL queries.	2
03	Write and execute advanced SQL queries using joins, subqueries, views, triggers, and window functions.	2
04	Normalize a given database up to BCNF and explain the effect of normalization on data redundancy.	2
05	Demonstrate transaction control and concurrency using SQL statements and recovery techniques.	2
06	Analyze query optimization and indexing techniques using EXPLAIN or ANALYZE commands.	2
07	Implement CRUD operations in a NoSQL database (e.g., MongoDB) and compare with SQL equivalents.	2
08	Develop a mini project integrating SQL and NoSQL databases for a practical application scenario.	2

Assessment:

Term Work: Term Work shall consist of at least 10 Experiments based on the above list. Also, Term work Journal must include at least 2 assignments.

Distribution of Marks for Term Work (25 Marks):

Laboratory Work 15 Marks

Assignments 05 Marks

Attendance 05 Marks

End Semester Practical/Oral Examination:

1. Pair of Internal and External Examiner should conduct practical / viva based on contents

2. Distribution of marks for practical/viva examination shall be as follows:

a. Practical performance 15 Marks

b. Viva 10 Marks

3. Evaluation of practical examination to be done based on the experiment performed and the output of the experiment during practical examination

4. Students work along with evaluation report to be preserved till the next examination

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20361172	Industrial Robotics Lab	--	02	-	--	01	-	01

Course Code	Course Name	Examination Scheme						
		Theory Marks				Term Work	Practical/ Oral	Total
		Internal assessment			End Sem. Exam			
		Test 1	Test 2	Avg. of 2 Tests				
20361172	Industrial Robotics Lab	--	--	--	--	25	25	50

Course Objectives:

1. Apply robot programming skills (teach pendant / offline programming) to execute point-to-point and continuous paths.
2. Design and implement simple grippers and assess gripping force requirements.
3. Analyze and implement trajectory tracking and basic force/impedance control experiments.
4. Integrate sensors (proximity, encoders, force/torque, vision) with industrial robots for pick-and-place tasks.
5. Demonstrate operation and basic design principles of material handling equipment (conveyors, industrial trucks, AGV mockup, ASRS demo).
6. Work in small teams to plan, execute, document, and present a mini project solving an industrially-relevant automation problem.

Course Outcomes:

1. To program industrial robot for point-to-point and path trajectories and verify accuracy.
2. To design, fabricate (basic) and test a gripper for a specified part and compute gripping forces.
3. To implement and evaluate closed-loop trajectory following and PD/Computed-Torque concepts (simulation or bench).
4. To integrate sensors and implement pick-place with object detection (sensor/vision).
5. To demonstrate key material handling systems (conveyor, palletizing, AGV basics) and select suitable equipment for a simple material flow problem.
6. To create a mini project, develop a lab report, and present findings demonstrating teamwork, documentation, and problem solving.

List of Experiments.

Sr No	List of Experiments	Hrs	LO Mapping
01	Study of industrial robot safety, robot anatomy, coordinate systems and teach pendant operations	02	LO1
02	Programming of industrial robot for point-to-point (PTP) motion and repeatability test.	02	LO1
03	Programming of industrial robot for continuous path (linear/circular interpolation) motion	02	LO1
04	Trajectory generation and trajectory tracking of robotic manipulator (simulation / hardware)	02	LO3
05	Implementation of PD / computed-torque based control for robot joint motion (simulation / demonstration)	02	LO3
06	Design and force analysis of a robotic gripper for given object geometry and payload	02	LO2
07	Integration of gripper/end-effector with industrial robot and pick-and-place operation	02	LO1, LO2
08	Sensor integration (proximity/limit/encoder) for condition-based robot operation	02	LO4
09	Vision-based object detection and robot pick-and-place operation (basic visual servoing demo)	02	LO4
10	Study and operation of conveyor system with robotic pick-and-place / palletizing	02	LO5
11	Study and demonstration of Automated Guided Vehicle (AGV) or mobile robot for material handling	02	LO5
12	Study and demonstration of Automated Storage and Retrieval System (ASRS) / RFID or barcode-based material tracking	02	LO5
13	Mini Project: Design and implementation of an integrated industrial robotic and material handling system	02	LO6

Sr No	List of Assignments / Tutorials	Hrs
01	Assignment covers the topics from Module 1	2
02	Assignment covers the topics from Module 2	2
03	Assignment covers the topics from Module 3	2
04	Assignment covers the topics from Module 4	2
05	Assignment covers the topics from Module 5	2
06	Assignment covers the topics from Module 6	2

Assessment:

Term Work: Term Work shall consist of at least 10 Experiments based on the above list. Also, Term work Journal must include at least 2 assignments.

Distribution of Marks for Term Work (25 Marks):

Laboratory Work 15 Marks

Assignments 05 Marks

Attendance 05 Marks

End Semester Practical/Oral Examination:

1. Pair of Internal and External Examiner should conduct practical / viva based on contents
2. Distribution of marks for practical/viva examination shall be as follows:
 - a. Practical performance 15 Marks
 - b. Viva 10 Marks
3. Evaluation of practical examination to be done based on the experiment performed and the output of the experiment during practical examination
4. Students work along with evaluation report to be preserved till the next examination

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20361173	IoT (Lab)	-	2	-	-	1	-	1

Course Code	Course Name	Examination Scheme						
		Theory Marks				Term Work	Practical/ Oral	Total
		Internal assessment			End Sem. Exam			
		Test 1	Test 2	Avg. of 2 Tests				
20361173	IoT (Lab)	--	--	--	--	25	25	50

Course Objectives:

1. To understand the hardware and software components of IoT systems, including microcontrollers, sensors, and communication protocols
2. To design and implement IoT applications using microcontrollers such as Node MCU and ESP32 with various sensors and communication interfaces.
3. To understand communication protocols such as HTTP, MQTT and CoAP used in IoT.
4. To simulate and analyze IoT network protocols such as RPL and 6LoWPAN using Cooja simulator.
5. To connect IoT devices to cloud platforms using web hook and IFTTT for data logging and automation
6. To evaluate IoT system performance and security aspects, including edge vs cloud computing and packet-level data analysis.

Course Outcomes:

1. Students will be able to interface sensors and actuators with IoT hardware using GPIOs and timers.
2. Students will be able to create basic web servers and dashboards using Node MCU/ESP32 to visualize IoT data.
3. Students will implement MQTT and CoAP communication protocols to establish device-to-device and Restful communications.
4. Students will use different simulator to model and analyze low-power IoT networks and protocol stacks.
5. Students will integrate IoT devices with cloud platforms using APIs and service like IFTTT.
6. Students will assess IoT system architecture, simulations and analysis.

Prerequisite: Engineering basics of Measurement

List of Experiments:

Sr No	List of Experiments	CO Mapping	Hrs
01	Blinking LED with Timers and GPIO using Node MCU/ESP32	LO 1	2
02	Creating a Web Server on Node MCU/ESP32 to Display Dummy Sensor Data	LO 2	2
03	MQTT-based Communication Between Two Devices	LO 3	2
04	LoRa Peer-to-Peer Communication for IoT Data Exchange	LO 3	2
05	Cooja Simulation: RPL Protocol Implementation	LO 4	2
06	Cooja Simulation: 6LoWPAN Packet Transmission and Fragmentation	LO 4	2
07	Implementing CoAP Client and Server using Python	LO 3	2
08	Wi-Fi Signal Strength Logger with ESP32	LO 1	2
09	Cloud Logging with Google Sheets using IFTTT/Web hooks	LO 5	2
10	IoT Security: Sniffing and Analyzing MQTT/HTTP Packets using Wire shark	LO 3	2
11	Edge vs Cloud Processing Latency Comparison	LO 6	2
12	Simulating 3-Layer IoT Architecture (Device-Edge-Cloud)	LO 6	2

Sr No	List of Assignments / Tutorials	Hrs
01	Assignment covers the topics from Module 1	2
02	Assignment covers the topics from Module 2	2
03	Assignment covers the topics from Module 3	2
04	Assignment covers the topics from Module 4	2
05	Assignment covers the topics from Module 5	2
06	Assignment covers the topics from Module 6	2

Assessment:

Term Work: Term Work shall consist of at least 10 Experiments based on the above list. Also, Term work Journal must include at least 2 assignments.

Distribution of Marks for Term Work (25 Marks):

Laboratory Work 15 Marks

Assignments 05 Marks

Attendance 05 Marks

End Semester Practical/Oral Examination:

1. Pair of Internal and External Examiner should conduct practical / viva based on contents

2. Distribution of marks for practical/viva examination shall be as follows:

a. Practical performance 15 Marks

b. Viva 10 Marks

3. Evaluation of practical examination to be done based on the experiment performed and the output of the experiment during practical examination

4. Students work along with evaluation report to be preserved till the next examination

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20361181	Biomedical Instrumentation Lab	-	2	-	-	2	-	2

Course Code	Course Name	Examination Scheme						
		Theory Marks				Term Work	Practical/ Oral	Total
		Internal assessment			End Sem. Exam			
		Test 1	Test 2	Avg. of 2 Tests				
20361181	Biomedical Instrumentation Lab	--	--	--	--	25	25	50

Course Objectives:

1. To identify various electrodes.
2. To analyze the sphygmomanometer.
3. To operate and demonstrate MRI, CT scan, and ultrasonography systems.
4. To analyze the DC defibrillator system.
5. To operate the sphygmomanometer.
6. To analyze the working of a pacemaker.

Course Outcomes:

1. Analyze EEG, EMG and ECG
2. Analyze response of audiometer
3. Analyze operation of MRI, CT Scan
4. Analyze working of Pacemaker
5. Analyze working of audiometer
6. Analyze operation of Ultrasonography machine

List of Experiments:

Sr No	List of Experiments	LO	Hrs
01	To study different electrodes for various application.	LO1	02
02	Calibrate & maintain ECG machine	LO1	02
03	Demonstrate the performance of EMG machine	LO1	02
04	Demonstrate the performance of EEG machine	LO1	02
05	To observe the functioning of DC defibrillator system and voltage waveforms	LO1	02
06	To observe response of human ear with audiometer	LO2	02

07	To measure blood pressure by sphygmomanometer	LO2	02
08	To plot study spirogram using spirometer.	LO2	02
09	To study working of pacemaker and various waveforms	LO4	02
10	To Demonstrate the operation of MRI	LO3	02
11	To Demonstrate the operation of Ultrasonography	LO6	02
12	To Demonstrate the operation of CT scan	LO3	02

Note:

- Minimum 8 experiments can be performed during semester for term work.
- Hospital visit is advised to understand the concept of Biomedical Instrumentation subject.

Sr No	List of Assignments / Tutorials	Hrs
01	Assignment covers the topics from Module 1	2
02	Assignment covers the topics from Module 2	2
03	Assignment covers the topics from Module 3	2
04	Assignment covers the topics from Module 4	2
05	Assignment covers the topics from Module 5	2
06	Assignment covers the topics from Module 6	2

Assessment:

Term Work: Term Work shall consist of at least 10 Experiments based on the above list. Also, Term work Journal must include at least 2 assignments.

Distribution of Marks for Term Work (25 Marks):

Laboratory Work	15 Marks
Assignments	05 Marks
Attendance	05 Marks

End Semester Practical/Oral Examination:

1. Pair of Internal and External Examiner should conduct practical / viva based on contents
2. Distribution of marks for practical/viva examination shall be as follows:
 - a. Practical performance 15 Marks
 - b. Viva 10 Marks
3. Evaluation of practical examination to be done based on the experiment performed and the output of the experiment during practical examination
4. Students work along with evaluation report to be preserved till the next examination

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20361182	Micro Electro Mechanical System Lab	--	02	-	--	01	-	01

Course Code	Course Name	Examination Scheme						
		Theory Marks				Term Work	Practical/ Oral	Total
		Internal assessment			End Sem. Exam			
		Test 1	Test 2	Avg. of 2 Tests				
20361182	Micro Electro Mechanical System Lab	--	--	--	--	25	25	50

Course Objectives:

1. Demonstrate MEMS device structures and fabrication concepts through simulations and case-based experiments.
2. Implement modeling and simulation of MEMS structures such as cantilever beams, sensors, and micro-actuators.
3. Analyze the behavior and performance characteristics of MEMS devices using simulation tools.
4. Interpret simulation and experimental results for MEMS sensors and actuators used in engineering applications.

Course Outcomes:

1. Explain the structure and working principles of MEMS devices and microsystems.
2. Demonstrate MEMS fabrication processes such as lithography, micromachining, and thin-film deposition through simulation or virtual tools.
3. Implement simulation models of MEMS structures such as cantilever beams and microstructures.
4. Analyze the performance of MEMS-based sensors used in automation and robotic systems.
5. Analyze the operating characteristics of MEMS micro-actuators.
6. Evaluate MEMS device performance through simulation or case-based applications.

List of Experiments.

Sr No	List of Experiments	Hrs	LO Mapping
01	Study of MEMS devices and microsystem components used in engineering applications	02	LO1
02	Study of MEMS fabrication processes such as lithography, deposition, and etching using virtual lab or multimedia resources	02	LO2
03	Modeling and simulation of MEMS cantilever beam and analysis of displacement and stress	02	LO3

04	Simulation of MEMS pressure sensor and analysis of sensor characteristics	02	LO4
05	Simulation of MEMS accelerometer used in robotic motion sensing	02	LO4
06	Simulation of electrostatic MEMS micro-actuator	02	LO5
07	Simulation of piezoelectric MEMS micro-actuator	02	LO5
08	Simulation of thermal MEMS micro-actuator	02	LO5
09	Case study and analysis of MEMS sensors used in robotics or automation systems	02	LO6
10	Mini-project: Design or analysis of a MEMS device using simulation tools	02 to 04	LO6

Sr No	List of Assignments / Tutorials	Hrs
01	Assignment covers the topics from Module 1	2
02	Assignment covers the topics from Module 2	2
03	Assignment covers the topics from Module 3	2
04	Assignment covers the topics from Module 4	2
05	Assignment covers the topics from Module 5	2
06	Assignment covers the topics from Module 6	2

Assessment:

Term Work: Term Work shall consist of at least 10 Experiments based on the above list. Also, Term work Journal must include at least 2 assignments.

Distribution of Marks for Term Work (25 Marks):

Laboratory Work 15 Marks

Assignments 05 Marks

Attendance 05 Marks

End Semester Practical/Oral Examination:

1. Pair of Internal and External Examiner should conduct practical / viva based on contents
2. Distribution of marks for practical/viva examination shall be as follows:
 - a. Practical performance 15 Marks
 - b. Viva 10 Marks
3. Evaluation of practical examination to be done based on the experiment performed and the output of the experiment during practical examination
4. Students work along with evaluation report to be preserved till the next examination

Course Code	Course Name	Examination Scheme						
		Theory Marks				Term Work	Practical/ Oral	Total
		Internal assessment			End Sem. Exam			
		Test 1	Test 2	Avg. of 2 Tests				
20361183	Hydraulics and Pneumatics Lab	--	--	--	--	25	25	50

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
20361183	Hydraulics and Pneumatics Lab	--	02	-	--	01	-	01

Course Objectives:

1. To understand basic principles of hydraulic and pneumatic systems.
2. To study working of pumps, actuators, and control valves.
3. To design and analyze basic hydraulic circuits.
4. To understand electro-hydraulic and electro-pneumatic systems.
5. To study pneumatic components, logic circuits, and air supply systems.
6. To develop skills for interpreting and troubleshooting fluid power systems.

Course Outcomes:

1. Apply basic principles of fluid power systems and fluid properties.
2. Demonstrate working of hydraulic pumps and actuators.
3. Design and interpret basic hydraulic circuits using control valves.
4. Analyze electro-hydraulic control systems.
5. Design basic pneumatic logic and control circuits.
6. Troubleshoot electro-pneumatic systems used in automation.

Sr No	List of Experiments	Hrs	LO Mapping
01	Study of basic fluid power systems and identification of hydraulic and pneumatic components	02	LO1
02	Study of physical properties of fluids and demonstration of Pascal's law using hydraulic setup	02	LO1
03	Study of hydraulic pumps (gear, vane, piston) and analysis of their construction and working	02	LO2
04	Study of hydraulic actuators (single acting and double acting cylinders) and performance analysis	02	LO2
05	Study and interpretation of hydraulic circuit symbols (ISO standard) and directional control valves	02	LO3

06	Design and analysis of basic hydraulic circuits (single and double acting cylinder operation)	02	LO3
07	Study of speed control circuits using meter-in and meter-out flow control methods	02	LO3
08	Study of electro-hydraulic components such as solenoid valves, relays, and sensors	02	LO4
09	Implementation of simple electro-hydraulic relay-based control circuit	02	LO4
10	Study of pneumatic system components including compressor and FRL unit	02	LO5
11	Study of pneumatic actuators and directional control valves used in automation	02	LO5
12	Design of pneumatic logic circuits using AND, OR, and time delay valves	02	LO5
13	Design and simulation of sequential pneumatic circuits using cascade method	02	LO6
14	Study of electro-pneumatic control system using solenoid valves and timers	02	LO6
15	Troubleshooting and fault analysis of pneumatic and electro-pneumatic systems	02	LO6

Sr No	List of Assignments / Tutorials	Hrs
01	Assignment covers the topics from Module 1	2
02	Assignment covers the topics from Module 2	2
03	Assignment covers the topics from Module 3	2
04	Assignment covers the topics from Module 4	2
05	Assignment covers the topics from Module 5	2
06	Assignment covers the topics from Module 6	2

Assessment:

Term Work:

Distribution of Marks for Term Work

Laboratory Work 15 Marks

Assignments 05 Marks

Attendance 05 Marks

End Semester Practical/Oral Examination:

1. Pair of Internal and External Examiner should conduct practical / viva based on contents
2. Distribution of marks for practical/viva examination shall be as follows:
 - a. Practical performance 15 Marks
 - b. Viva 10 Marks
3. Evaluation of practical examination to be done based on the experiment performed and the output of the experiment during practical examination
4. Students work along with evaluation report to be preserved till the next examination

Vertical -4

VSEC

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2036411	3-D Printing Lab (Mini Project)	--	2*+2	-	--	2	-	2

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
2036411	3-D Printing Lab (Mini Project)	-	-	-	-	-	50	25	75

Rational:

Its ability to create physical objects from digital designs, driven by advantages like rapid prototyping, design freedom, cost-efficiency, and customization. It allows for faster product development cycles, reduces material waste, and enables the creation of complex, lightweight, or personalized parts that are difficult or impossible to produce with traditional manufacturing.

Course Objectives:

1. To familiarize students with part programming for tool path generation for machining operations, using G-M codes.
2. To make students acquaint with developing tool path from CAD to CAM system, using computer aided tool.
3. To prepare students for developing a 3D printed part using, FDM technique from CAD data.
4. To enable students to know the process of interpreting and processing medical scan data.
5. To introduce large-scale robotic 3D printing systems, FGF technology, material handling, and industrial applications of robotic additive manufacturing.
6. To study defect analysis, dimensional inspection, troubleshooting methods, sustainability aspects, and emerging industrial and biomedical applications of additive manufacturing.

Course Outcomes:

1. Model engineering components using CAD tools and prepare them for additive manufacturing by generating STL files and slicing with appropriate process parameters..
2. Perform 3D printing operations with different materials and machines by varying key process parameters and analyse their effect on build quality and efficiency.
3. Apply reverse engineering techniques using 3D scanning and evaluate printed components for dimensional accuracy and surface defects.

4. Able to apply appropriate Tools and Techniques to understand and analyze the problems following professional ethics with focus on societal and environmental aspects.
5. Work as a team and communicate results in an effective way.
6. Make decisions as an individual or as team member to manage tasks and also Engage in independent and life-long learning with ability to adapt to new and technological changes.

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Basics of engineering graphics, CAD modelling, manufacturing processes, and computer fundamentals.	00	--
I	Introduction to 3D Printing and CAD Modelling	Introduction to additive manufacturing, types of 3D printing technologies, applications of 3D printing, CAD modelling of engineering components, conversion of CAD models into STL format, basics of slicing software.	04	CO1
II	STL Slicing and Process Parameters	Slicing of STL files, generation of G-code, study of process parameters such as layer thickness, build orientation, infill density, print speed, and support structures. Effect of process parameters on build time and quality.	04	CO2
III	FDM 3D Printing Operations	Principle and working of FDM 3D printer, 3D printing of components by varying layer thickness, orientation, and infill density. Analysis of dimensional accuracy and build quality.	05	CO2 & CO3
IV	Reverse Engineering and 3D Scanning	Fundamentals of reverse engineering, working of 3D scanner, scanning of real-life objects, point cloud generation, CAD reconstruction of scanned objects, preparation of scanned models for printing.	04	CO3 & CO4
V	Large Scale Robotic 3D Printing	Introduction to FGF large-scale robotic 3D printing, robotic additive manufacturing systems, 3D printing using PETG material, material handling, process workflow, industrial applications of robotic 3D printing.	04	CO5
VI	Defect Analysis and Industrial Applications	Identification and analysis of defects in 3D printed components, dimensional inspection, surface finish analysis, troubleshooting methods, biomedical and industrial applications of additive manufacturing, sustainability and future trends in 3D printing.	04	CO6

Text Books

1. Ian Gibson, David Rosen and Brent Stucker, —Additive Manufacturing TechnologiesI, 2nd Edition, Springer Publication, 2015.
2. A. K. Choudhary and J. P. Davim, —3D Printing and Additive Manufacturing TechnologiesI, 1st Edition, Springer Publication, 2019.
3. S. B. Awari, V. R. Thorat, S. R. Ambade and A. V. Kothari, —Additive Manufacturing and 3D Printing Technology: Principles and ApplicationsI, 1st Edition, CRC Press, 2021.
4. P. N. Rao, —CAD/CAM: Principles and ApplicationsI, 3rd Edition, Tata McGraw Hill Publication, 2010.
5. A. K. Jain, —Advanced Manufacturing ProcessesI, 1st Edition, Khanna Publishers, New Delhi, 2009.

Reference Books

1. Ibrahim H. Mondal and Md. Enamul Hoque, —Theory and Practice of Additive Manufacturing, 1st Edition, Wiley Publication, 2023.
2. Debasis Pradhan, Rajesh Nayak and Dushmanta Kumar Thakur, —Fundamentals and Applications of Additive Manufacturing, 1st Edition, CRC Press, 2025.
3. Hod Lipson and Melba Kurman, —Fabricated: The New World of 3D Printing, 1st Edition, Wiley Publication, 2013.
4. P. C. Sharma, —Production Engineering, 7th Edition, S. Chand Publication, New Delhi, 2014.
5. Mikell P. Groover, —Automation, Production Systems and Computer Integrated Manufacturing, 4th Edition, Pearson Education, 2015.

List of Experiments.

Sr No	List of Experiments	LO Mapping	Hrs
01	Study of different 3D printing technologies and components of FDM printer.	LO1	02
02	Modelling of Engineering component and conversion of STL format.	LO1	02
03	Slicing of STL file and study of effect of process parameter like layer thickness, orientation, and infill on build time using software a. Exercise 1: Component-1 b. Exercise 2: Component-2	LO2	02
04	3D printing of modelled component by varying layer thickness.	LO3	02
05	3D printing of modelled component by varying orientation.	LO4	02
06	3D printing of modelled component by varying infill density	LO5	02
07	Modelling of component using 3D Scanner of real-life object of unknown dimension in reverse engineering.	LO5	02
08	3D printing of above modelled component.	LO6	02
09	3D printing the component using FGF large scale robotic 3D printer.	LO6	02
10	3D printing the component using PETG material in large scale robotic 3D printer.	LO6	02
11	Identifying the defects in 3D Printed components	LO6	02
12	Dimensional accuracy and surface finish measurement of 3D printed parts.	LO6	02

Assessment:

Project work: A working model or a simulation model or a study report leading to a conclusion as anticipated in semester –VI is required to be used for awarding marks. A proper rubric should be framed.

Note: Students shall form a group of 3 to 4 students, while forming a group shall not be allowed less than three or more than four students, as it is a group activity.

Term Work: Term Work shall consist of at least 10 Experiments based on the above list.

Distribution of Marks for Term Work (50 Marks):

Laboratory Work (Experiment)	15 Marks
Mini Project Report & Presentation	15 Marks
Model of Project	15 Marks
Attendance	05 Marks

End Semester Practical/Oral Examination:

1. Pair of Internal and External Examiner should conduct practical / viva based on contents

2. Guideline for assessment of Mini Project- Practical/ Oral Examination: 25 Marks

Report	05 Marks
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Working Model	10 Marks
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Presentation & Viva	10 Marks
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3. Evaluation of practical examination to be done based on the experiment performed and the output of the experiment during practical examination

4. Students work along with evaluation report to be preserved till the next examination

Vertical -5

AEC

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
2036511	Professional Skills	--	2*+2	-	--	2	-	2

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment			End Sem Exam	Exam Duration (in Hrs)			
		Test 1	Test 2	Total					
2036111	Professional Skills	-	-	-	-	-	25	-	25

Course Rationale:

This course equips engineering students with professional communication, ethical values and soft skills essential for global industry demands. It enhances technical writing, presentation and interpersonal abilities through practical sessions, fostering confidence, teamwork and social responsibility as competent engineering professionals.

Course Objectives:

1. Develop effective speaking, presentation, group discussion and interview skills required for employability and professional success.
2. Analyse and evaluate AI-generated content for accuracy, bias, credibility and ethical professional use.
3. Prepare clear, concise and professional technical/business documents such as resumes, reports, SOP and formal correspondence.
4. Demonstrate managerial and collaborative skills required for conducting meetings, preparing proposals and workplace communication.
5. Apply interpersonal, critical thinking, negotiations, time management, leadership, decision-making and adaptability skills for lifelong learning and career growth.
6. Exhibit ethical values, intellectual property awareness, digital responsibility and professionalism in workplace practices.

Course Outcomes -

1. Demonstrate effective speaking skills through participation in group discussions, interviews, presentations and collaborative activities.
2. Analyse and evaluate AI-generated content in terms of quality, bias, reliability and ethical use for academic and professional purposes.
3. Create professional documents such as resumes, cover letters, statements of purpose, reports and other technical/business correspondence using standard formats.
4. Organise and execute professional meetings, proposals and technical papers using appropriate workplace communication strategies and digital tools.
5. Apply and demonstrate interpersonal skills such as leadership, critical thinking, negotiation, decision-making, time management and adaptability to support career growth and lifelong learning.
6. Exhibit ethical practices, digital responsibility and awareness of intellectual property rights in professional environments.

DETAILED SYLLABUS

Sr. No.	Name of the Module	Content	Hours	CO Mapping
I	Employability Skills: Advanced Speaking	1.1. Group Discussions (GDs) - Purpose, Evaluation parameters, GD etiquette - Types: Factual (Regular & Controversial), Abstract, Case-based (Textual & Picture-based) and Role-plays (Textual & Cue-card-based) - Online GDs and digital collaboration etiquette (Zoom, Google-Meet, MS Teams etc.) 1.2. Personal Interviews - Planning and preparation for Technical and HR interviews - Interview types: Structured, Stress, Behavioural and Case-based - Virtual interviews, AI-assisted interview simulations and post-interview reflections (Google Interview Warmup, Yoodli, Pramp etc.) 1.3. Presentation Strategies - Purpose, Audience analysis, Content organisation - Slide design, Text, Infographics and AI-based presentation design (Canva, Beautiful.ai, Gamma AI etc.) - Division of roles, collaborative preparation and seamless transitions	8	CO1
II	Critical AI Literacy	- Nature of AI Generated Language - Understanding AI-generated language as probabilistic output and distinguishing between Fluency and Accuracy while identifying risks such as AI Slop and Hallucination - Evaluating AI Content - Differentiating between Evidence and Assertion and applying verification strategies to assess Accuracy, Clarity and Depth - Bias, Framing and Information Silos - Understanding linguistic Bias, Framing, Information Silos and Selective Exposure that shape AI responses and limit perspective - Drafting, Editing and Prompt Engineering with AI - Using AI for Drafting and Editing through effective Prompt Engineering while maintaining originality, specificity and professional voice - Identifying Low-Quality AI Content - Identify markers of low-quality AI content, such as repetition, lack of depth, and absence of concrete detail, while applying ethical practices that distinguish Assistance from Substitution and ensure responsible AI use	4	CO2
III	Employability Skills: Advanced	3.1. Cover Letter & Resume - Difference between Bio-data, Resume and CV - Resume types: Chronological, Functional and Hybrid	8	CO3

	Technical Writing	• Resume structure and content • Digital resumes, Application Tracking System (ATS) friendly formatting and LinkedIn portfolio integration 3.2. Reports • Purpose: Importance of reports in engineering and industry decision-making • Classification of Reports: Subject Matter - Technical, R&D, Operations, Finance, Marketing etc. Time Interval - Periodic, One-time, Special Function - Informational, Examinational and Analytical Reader Writer Relationship – Administrative, Professional and Independent Format – Memo, Letter, Short, Long, E-Reports, Digital Dashboards Digital and AI-aided reporting formats (Power BI, Google Data Studio, ChatGPT etc. for summaries) • Parts of a Long Formal Report Prefatory Parts: Title Fly, Title Page, Letter of Transmittal, Table of Contents, List of Illustrations, Executive Summary Report Proper (Main Body): Introduction, Intervening chapters with headings as per the topic, Result & Analysis, Summary (Informational) Or Summary & Conclusions (Examinational) Or Summary, Conclusions & Recommendations (Analytical) Appended Parts: Appendix, List of References, Glossary, Index • Language and Style of Reports Tense, person and voice appropriate to reports Numbering and labelling conventions for figures, tables, charts, equations etc. Referencing formats (APA, MLA etc.) Plagiarism checks (Turnitin, Grammarly, other AI-detectors etc.) 3.3. Statement of Purpose (SOP) • Purpose: SOPs for international internships, funding for research programs and higher education • Structure: Paragraph 1 - Hook (introducing yourself in a unique manner and demonstrating your passion for the field) Paragraph 2 - Academic Background Paragraph 3 - Research-oriented Endeavours Paragraph 4 - All-round Development Paragraph 5 - Reasons for choosing the Program and Institution Paragraph 6 - Conclusion and Thank-you note		
IV	Workplace Managerial Skills	4.1. Conducting Business Meetings • Types of Meetings: Executive Meetings - Decide	6	CO4

		Management Meetings - Implement Committee Meetings - Recommend Review Meetings - Evaluate Planning Meetings - Design the future Crisis Meetings - Resolve urgency • Roles and Responsibilities: Chair - Leads meeting, maintains order, ensures decisions Secretary - Records minutes, documents decisions, tracks actions Core Members - Discuss issues, contribute ideas, make decisions Specialists - Provide expertise, clarify issues, support decisions • Documentation: Notice, Agenda and Minutes • Meeting Platforms and AI Assistance: Virtual and hybrid meeting management, online etiquette, use of collaborative tools (Google Meet, Slack, Trello etc.) Digital documentation, e-Minutes and collaborative platforms like Notion or Microsoft OneNote 4.2. Writing Proposals • Types: Solicited vs. Unsolicited • Format: Memo, Short and Long • Parts of a Short Proposal: Purpose of the proposal Need for the proposal (SWOT analysis) Description of the plan (budget, timeline etc.) Evidence of ability to deliver Benefits to both parties Conclusion 4.3. Structuring and Formatting Technical Papers • Components: Title, Author(s) & Affiliations, Abstract, Keywords/Index Terms, Introduction, Sections with different headings, Results, Discussion, Conclusion, Acknowledgement, References, Biography • Formatting and referencing (IEEE style) • Use of AI tools for grammar check, plagiarism validation and submission to international conferences		
V	21st Century Professional Skills	5. Interpersonal Skills • Emotional Intelligence • Critical thinking, Reflection & Self Analysis • Leadership and Motivation Strategies • Conflict Management and Negotiation • Time Management • Anger and Stress Management Decision Making and Adaptability	6	CO5

VI	Digital Ecosystems and Ethics	6.1. Intellectual Property Rights (IPR) • Categories: Copyrights, Trademarks, Patents, Industrial Designs, Geographical Indications, Integrated Circuits and Trade Secrets • AI-generated content ownership, open-source ethics, digital rights management • Social Media Content and Legal Responsibility 6.2. Case Studies on Corporate and Engineering Ethics • Real-world ethical dilemmas • Corporate Social Responsibility (CSR) • Environmental Social Governance (ESG) • Sustainability and Green Engineering • AI Ethics in Design and Automation	4	CO6
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ASSESSMENTS

List of Compulsory Assignments for Term Work:

Module 1: GD documentation with Topic, Type and Highlights in 5 bullet points

Module 2: Assignment on Prompt Engineering

Module 3: **A** - Cover Letter, Resume and **B** - Statement of Purpose

Module 4: Meeting Documentation (Notice, Agenda & Minutes)

Module 5: Activities on Interpersonal Skills

Module 6: Case study on Ethical Dilemma

(To be conducted in the form of short notes, quizzes, role plays, case studies, presentations or MCQs)

SUGGESTED ASSIGNMENTS

Module No. & Title	Assignments and Activities
Module 1 Employability Skills: Advanced Speaking	• Conduct 4-5 types of Group Discussions and attach the summary and photographs as proof of activity • Prepare group presentation slides, based on the report topic (Examinational or Analytical) using the latest presentation software (PowerPoint, Canva, Prezi etc.) and engage students in giving critical feedback • Present an 'Elevator Pitch' for either self-introduction or product or new business idea • Conduct mock interviews
Module 2 Critical AI Literacy	• Select an essay/article/video generated by AI (or give a general prompt) and analyse and detect AI bias (gender pronouns, gender-based occupational stereotypes, racial identifiers, historical bias etc.) • Differentiate between weak and strong prompts • Source verification task • AI versus human writing analysis

Module 3 Employability Skills: Advanced Technical Writing	• Draft a professional Cover Letter and Resume • Write a Statement of Purpose (SOP) for higher studies or internships • Develop or enhance a LinkedIn profile showcasing technical and project achievements • Prepare a book-format analytical/informational report using IEEE/APA/MLA referencing and plagiarism check tools (Grammarly/Turnitin). Main body should be minimum 20 pages excluding the Preparatory and Appended parts
Module 4 Workplace Managerial Skills	• Conduct a mock meeting with assigned roles (Chairperson, Secretary, Members) and generate documentation with the help of the latest software • Write a short proposal addressing an engineering or sustainability problem • A quiz on the formatting aspects of a Technical Paper
Module 5 21st Century Professional Skills	• Activities for developing team work, leadership, emotional intelligence etc. • Problems and Puzzles on Critical thinking • Case-studies or Role-plays or Questionnaires on Conflict Resolution, Negotiation Skills and Decision Making • Application of SWOT analysis • Eisenhower Matrix to analyse a daily time log
Module 6 Digital Ecosystems and Ethics	• Conduct an awareness quiz on IPR laws and ethical business practices. • Analyse a corporate ethics case study • Digital footprint audit of Self Profile • Fake News and Deep Fake verification task

IMPORTANT NOTE

1. The main project/book report should contain a minimum of 20 pages, excluding prefatory and appended matter.
2. Each project group must consist of a minimum of 4 and a maximum of 6 students.
3. End-semester presentation will be based on the book/project report.

GUIDELINES FOR TERM WORK MARKS DISTRIBUTION

Components	Marks 25	Description
6 Compulsory Assignments	5	Based on completion and quality of module-wise tasks.
Attendance & Participation	5	Class engagement, teamwork, and professionalism.

GD	5	Based on group and individual performance
Report Hardcopy	5	Quality, structure and originality of the report
Oral Report Presentation	5	Based on group and individual performance

RECOMMENDED RESOURCES:

1. Raman, M., & Sharma, S. (2022). Technical communication: Principles and practice (4th ed.). Oxford University Press.
2. Anderson, P. V. (2018). Technical communication: A reader-centred approach (9th ed.). Cengage Learning.
3. Lesikar, R. V., Flatley, M. E., & Rentz, K. (2008). Basic business communication (11th ed.). McGraw-Hill.
4. Institute of Electrical and Electronics Engineers. (2020). IEEE code of ethics. IEEE.
5. American Society of Mechanical Engineers. (2021). ASME code of ethics of engineers. ASME.
6. Guffey, M. E., & Loewy, D. (2018). Essentials of business communication (11th ed.). Cengage Learning.
7. Sharma, R. C., & Mohan, K. (2017). Business correspondence and report writing (4th ed.). Tata McGraw-Hill Education.
8. Ram, A. (2018). Place mentor: Tests of aptitude for placement readiness. Oxford University Press.
9. Kumar, S., & Lata, P. (2018). Communication skills: A workbook. Oxford University Press.
10. Butterfield, J. (2017). Verbal communication: Soft skills for a digital workplace. Cengage Learning.
11. Crawford, K. (2021). Atlas of AI: Power, politics, and the planetary costs of artificial intelligence. Yale University Press.
12. Kearns, M., & Roth, A. (2019). The ethical algorithm: The science of socially aware algorithm design. Oxford University Press.

ONLINE RESOIRCES

1. Ravichandran, T. (n.d.). *Enhancing soft skills and personality* [NPTEL course]. National Programme on Technology Enhanced Learning. <https://nptel.ac.in>
2. Ravichandran, T. (n.d.). *Developing soft skills and personality* [SWAYAM MOOC]. SWAYAM. <https://swayam.gov.in>
3. Coursera. (n.d.). *Business communication for success* [Online course]. <https://www.coursera.org>
4. edX. (n.d.). *Business communication for success* [Online course]. <https://www.edx.org>
5. All India Council for Technical Education. (n.d.). *Employability enhancement and life skills modules*. AICTE NEAT. <https://neat.aicte-india.org>
6. Indian Institute of Technology Guwahati. (n.d.). *Virtual English and communication virtual lab*. Virtual Labs. <https://ve-iitg.vlabs.ac.in/>
7. Institute of Technology Bombay. (n.d.). *Professional communication virtual lab*. Virtual Labs. <http://vlabs.iitb.ac.in/vlabs-dev/labs/communication/>
8. TCS iON. (n.d.). *Communication skills* [Online course]. <https://www.tcsion.com/courses/tcs-ion/communication-skills/>
9. TCS iON. (n.d.). *Introduction to soft skills* [Online course]. <https://www.tcsion.com/courses/interview-and-job-prep/introduction-to-soft-skills/>
10. TCS iON. (n.d.). *Presentation skills* [Online course]. <https://www.tcsion.com/courses/tcs-ion/presentation-skills/>

QUESTION PAPER PATTERN (External and Internal)

Letter Grades and Grade Points:

Semester GPA/ Programme CGPA Semester/ Programme	% of Marks	Alpha-Sign/ Letter Grade Result	Grading Point
9.00 - 10.00	90.0 – 100	O (Outstanding)	10
8.00 - < 9.00	80.0 - < 90.0	A+ (Excellent)	9
7.00 - < 8.00	70.0 - < 80.0	A (Very Good)	8
6.00 - < 7.00	60.0 - < 70.0	B+ (Good)	7
5.50 - < 6.00	55.0 - < 60.0	B (Above Average)	6
5.00 - < 5.50	50.0 - < 55.0	C (Average)	5
4.00 - < 5.00	40.0 - < 50.0	P (Pass)	4
Below 4.00	Below 40.0	F (Fail)	0
Ab (Absent)	-	Ab (Absent)	0

Sd/-

Dr. S.M.Khot
BoS-Chairman-Mechanical Engineering
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Sd/-

Dr. Deven Shah
Associate Dean
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Sd/-

Prof. Shivram S. Garje
Dean
Faculty of Science & Technology